1 You are given that $f(x) = (x + 2)^2(x - 3)$.

(i) Sketch the graph of $y = f(x)$.

[3]

(ii) State the values of x which satisfy $f(x + 3) = 0$.

[2]

2

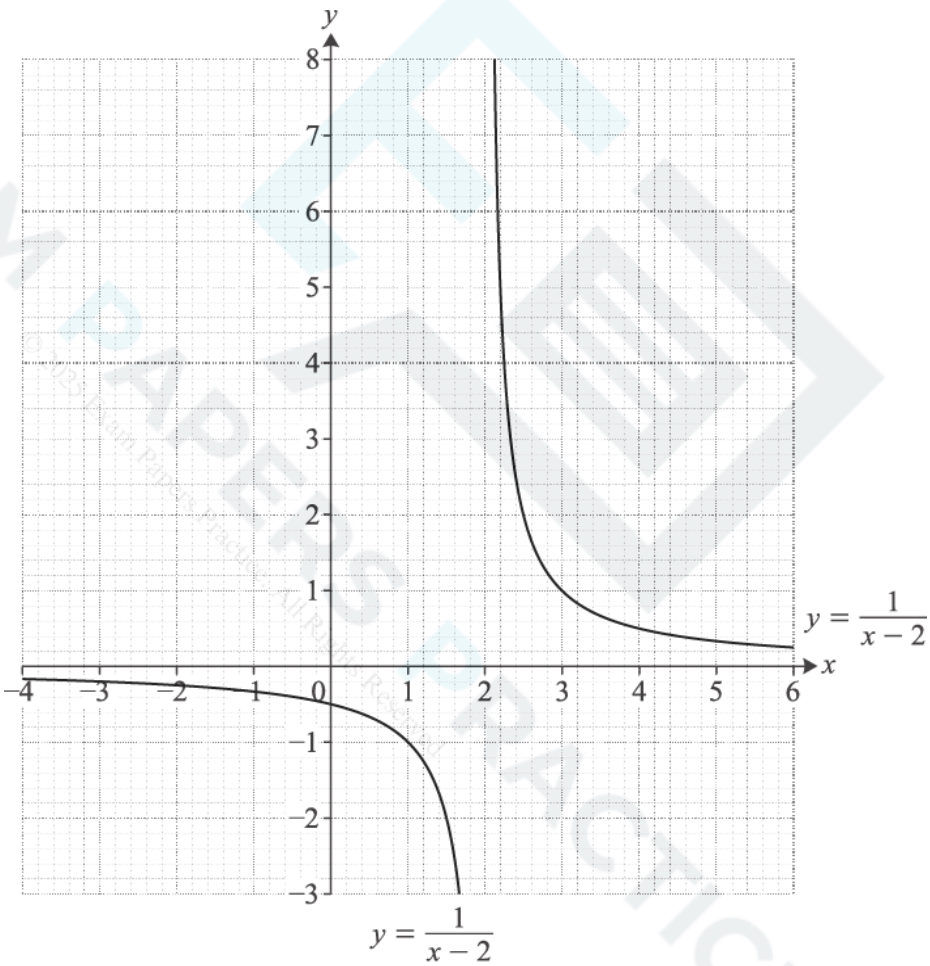


Fig. 12

Fig. 12 shows the graph of $y = \dfrac{1}{x-2}$.

(i) Draw accurately the graph of $y = 2x + 3$ on the copy of Fig. 12 and use it to estimate the coordinates of the points of intersection of $y = \frac{1}{x-2}$ and $y = 2x + 3$.

[3]

(ii) Show algebraically that the x-coordinates of the points of intersection of $y = \frac{1}{x-2}$ and $y = 2x + 3$ satisfy the equation $2x^2 - x - 7 = 0$. Hence find the exact values of the x-coordinates of the points of intersection.

[5]

(iii) Find the quadratic equation satisfied by the x-coordinates of the points of intersection of $y = \frac{1}{x-2}$ and $y = -x + k$. Hence find the exact values of k for which $y = -x + k$ is a tangent to $y = \frac{1}{x-2}$.

[4]

3 Express $3x^2 - 12x + 5$ in the form $a(x - b)^2 - c$. Hence state the minimum value of y on the curve $y = 3x^2 - 12x + 5$.

[5]

4 You are given that $f(x) = (2x - 3)(x + 2)(x + 4)$.

(i) Sketch the graph of $y = f(x)$.

[3]

(ii) State the roots of $f(x - 2) = 0$.

[2]

(iii) You are also given that $g(x) = f(x) + 15$.

A Show that $g(x) = 2x^3 + 9x^2 - 2x - 9$.

[2]

B Show that $g(1) = 0$ and hence factorise $g(x)$ completely.

[5]

5

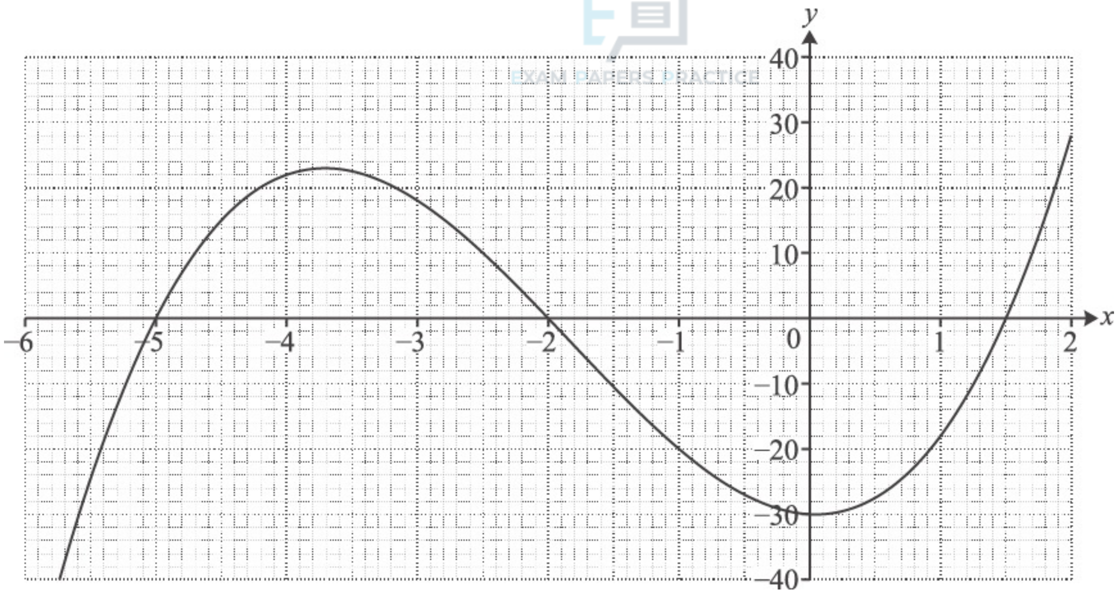


Fig. 12

Fig. 12 shows the graph of a cubic curve. It intersects the axes at (–5, 0), (–2, 0), (1.5, 0) and (0, –30).

(i) Use the intersections with both axes to express the equation of the curve in a factorised form.

[2]

(ii) Hence show that the equation of the curve may be written as $y = 2x^2 + 11x^2 - x - 30$.

[2]

(iii) Draw the line $y = 5x + 10$ accurately on the graph. The curve and this line intersect at (–2, 0); find graphically the x-coordinates of the other points of intersection.

[3]

(iv) Show algebraically that the x-coordinates of the other points of intersection satisfy the equation

$$2x^2 + 7x - 20 = 0.$$

Hence find the exact values of the x-coordinates of the other points of intersection.

[5]

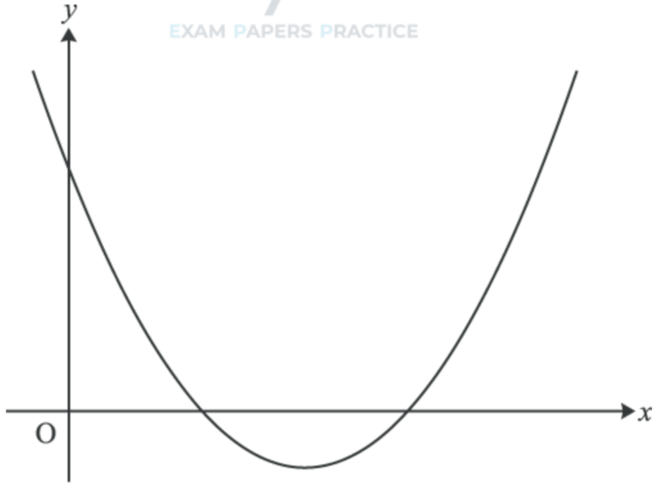


Fig. 11

Fig. 11 shows a sketch of the curve with equation $y = (x - 4)^2 - 3$.

(i) Write down the equation of the line of symmetry of the curve and the coordinates of the minimum point.

[2]

(ii) Find the coordinates of the points of intersection of the curve with the x-axis and the y-axis, using surds where necessary.

[4]

(iii) The curve is translated by $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Show that the equation of the translated curve may be written as $y = x^2 - 12x + 33$.

[2]

(iv) Show that the line $y = 8 - 2x$ meets the curve $y = x^2 - 12x + 33$ at just one point, and find the coordinates of this point.

[5]

7

(i) Find the coordinates of the points of intersection of the curve $y = 2x^2 - 5x - 3$ with the axes. **[3]**

(ii) Find the coordinates of the points of intersection of the curve y = $2x^2 - 5x - 3$ and the line $y = x + 3$. **[4]**

(iii) Find the set of values of k for which the line $y = x + k$ does not intersect the curve $y = 2x^2 - 5x - 3$. **[5]**

8

Fig. 9 shows the curves $y = \frac{1}{x+2}$ and $y = x^2 + 7x + 7$.

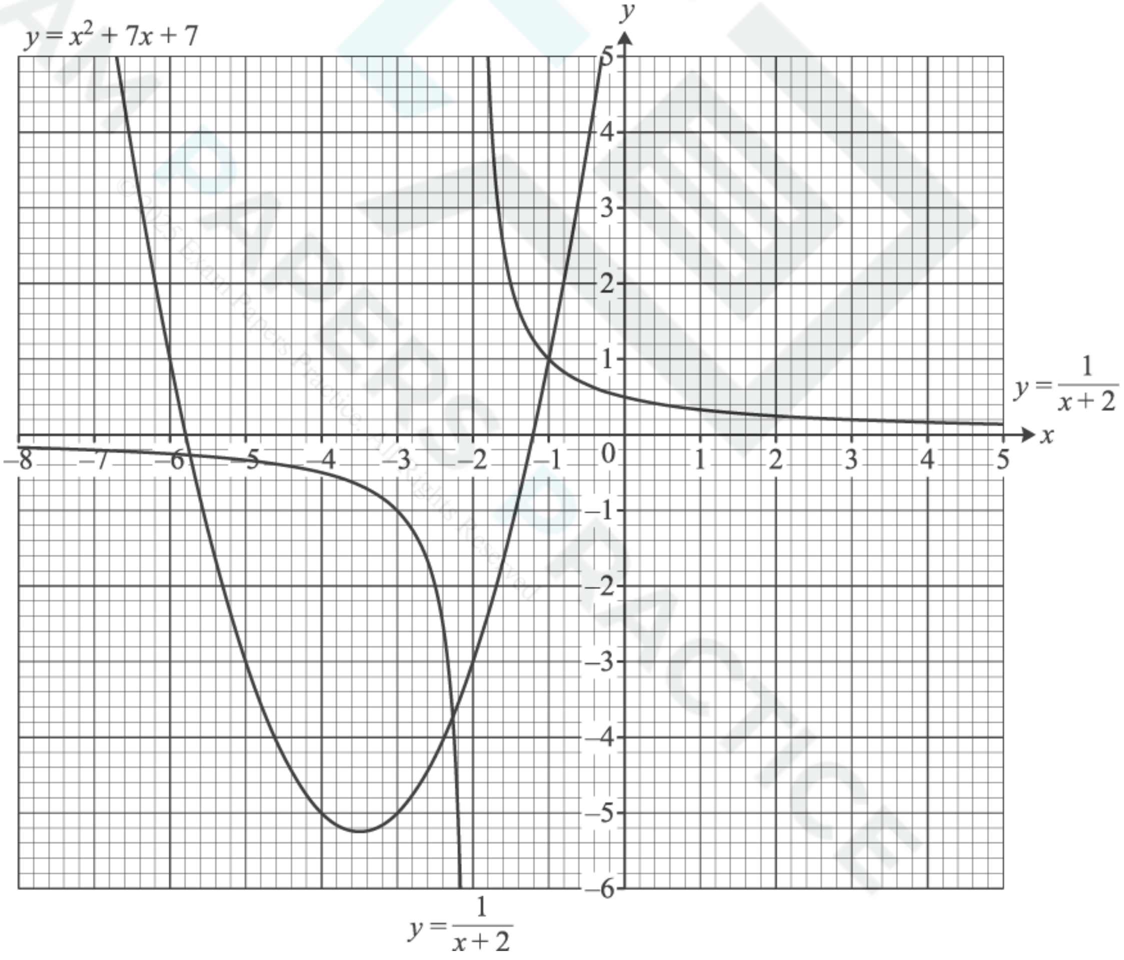


Fig.9

(i) Use Fig. 9 to estimate graphically the roots of the equation $\frac{1}{x+2} = x^2 + 7x + 7$.

[2]

(ii) Show that the equation in part (i) may be simplified to $x^3 + 9x^2 + 21x + 13 = 0$. Find algebraically the exact roots of this equation.

[7]

(iii) The curve $y = x^2 + 7x + 7$ is translated by $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

(*A*) Show graphically that the translated curve intersects the curve $y = \frac{1}{x+2}$ at only one point. Estimate the coordinates of this point.

[2]

(*B*) Find the equation of the translated curve, simplifying your answer.

[2]

9

(i) Solve the equation $(x-2)^2 = 9$.

[2]

(ii) Sketch the curve $y = (x-2)^2 - 9$, showing the coordinates of its intersections with the axes and its turning point.

[3]

(i) The point P (4, –2) lies on the curve $y = f(x)$. Find the coordinates of the image of P when the curve is transformed to $y = f(5x)$.

[2]

(ii) Describe fully a single transformation which maps the curve $y = \sin x°$ onto the curve $y = \sin(x - 90)°$.

[2]

11 Fig. 8 shows the graph of $y = g(x)$.

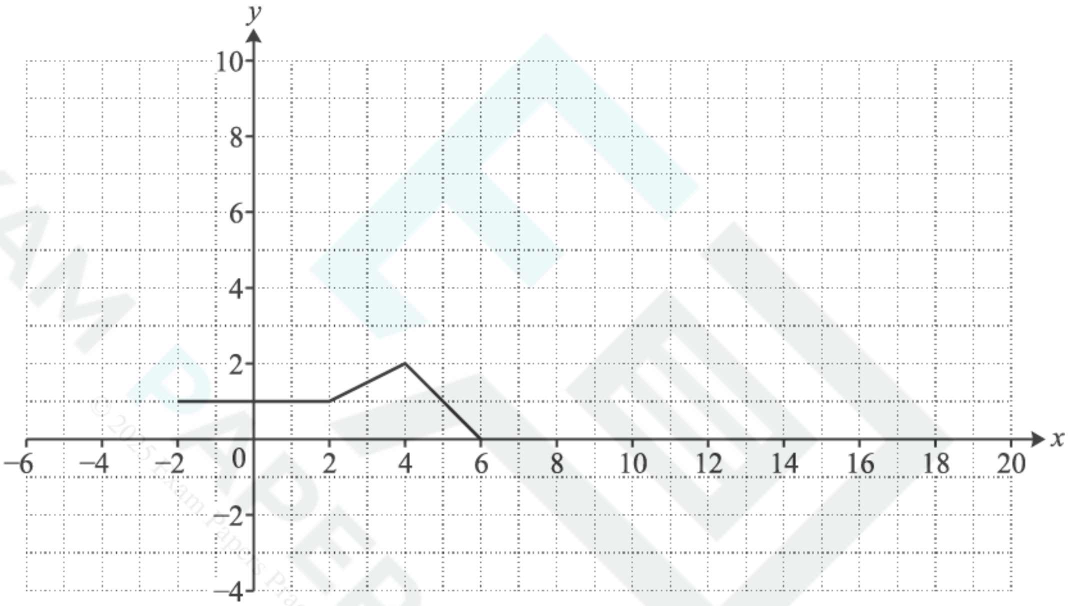


Fig. 8

Draw the graph of

(i) $y = g(2x)$,

[2]

(ii) $y = 3g(x)$.

[2]

12 The point R (6, –3) is on the curve $y = f(x)$.

(i) Find the coordinates of the image of R when the curve is transformed to $y = \frac{1}{2}f(x)$.

[2]

(ii) Find the coordinates of the image of R when the curve is transformed to $y = f(3x)$.

[2]

13 The gradient of a curve is given by $\frac{dy}{dx} = 4x + 3$. The curve passes through the point (2, 9).

(i) Find the equation of the tangent to the curve at the point (2, 9).

[3]

(ii) Find the equation of the curve and the coordinates of its points of intersection with the x-axis. Find also the coordinates of the minimum point of this curve.

[7]

(iii) Find the equation of the curve after it has been stretched parallel to the x-axis with scale factor $\frac{1}{2}$. Write down the coordinates of the minimum point of the transformed curve.

[3]

14

(i) On the same axes, sketch the curves $y = 3^x$ and $y = 3^{2x}$, identifying clearly which is which.

[3]

(ii) Given that $3^{2x} = 729$, find in either order the values of 3^x and x.

[2]

15

(i) Fig. 5 shows the graph of a sine function.

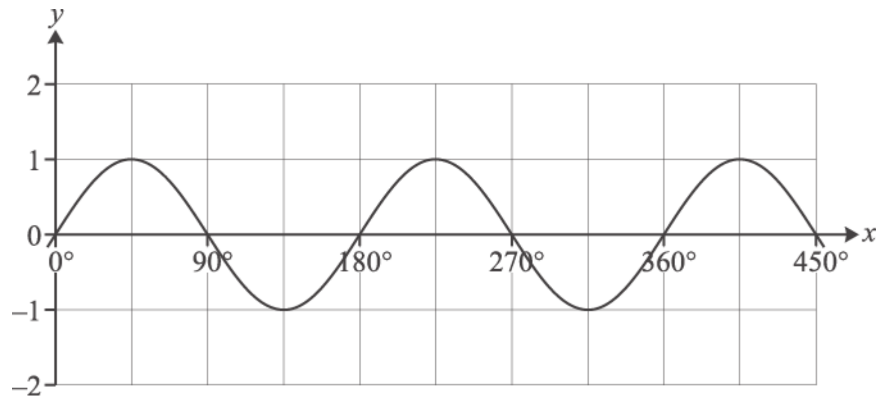


Fig. 5

State the equation of this curve. **[2]**

(ii) Sketch the graph of $y = \sin x - 3$ for $0° \leqslant x \leqslant 450°$. **[2]**

16

Fig. 8 shows parts of the curves $y = f(x)$ and $y = g(x)$, where $f(x) = \tan x$ and $g(x) = 1 + f(x - \frac{1}{4}\pi)$.

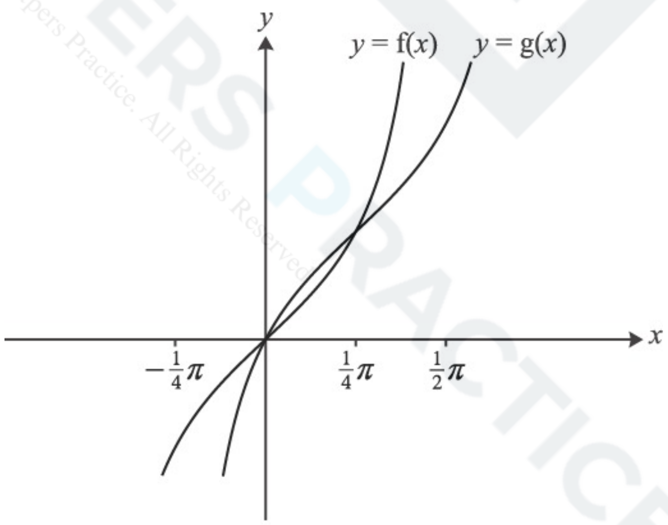


Fig. 8

(i) Describe a sequence of two transformations which maps the curve $y = \mathrm{f}(x)$ to the curve $y = \mathrm{g}(x)$.

[4]

It can be shown that $\mathrm{g}(x) = \dfrac{2\sin x}{\sin x + \cos x}$.

(ii) Show that $\mathrm{g}'(x) = \dfrac{2}{(\sin x + \cos x)^2}$. Hence verify that the gradient of $y = \mathrm{g}(x)$ at the point $(\frac{1}{4}\pi, 1)$ is the same as that of $y = \mathrm{f}(x)$ at the origin.

[7]

(iii) By writing $\tan x = \dfrac{\sin x}{\cos x}$ and using the substitution $u = \cos x$, show that $\displaystyle\int_0^{\frac{1}{4}\pi} \mathrm{f}(x)\,\mathrm{d}x = \int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{u}\,\mathrm{d}u$.

Evaluate this integral exactly.

[4]

(iv) Hence find the exact area of the region enclosed by the curve $y = \mathrm{g}(x)$, the x-axis and the lines

$x = \frac{1}{4}\pi$ and $x = \frac{1}{2}\pi$.

[2]

17 Fig. 8 shows the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = (1 - x)\mathrm{e}^{2x}$, with its turning point P.

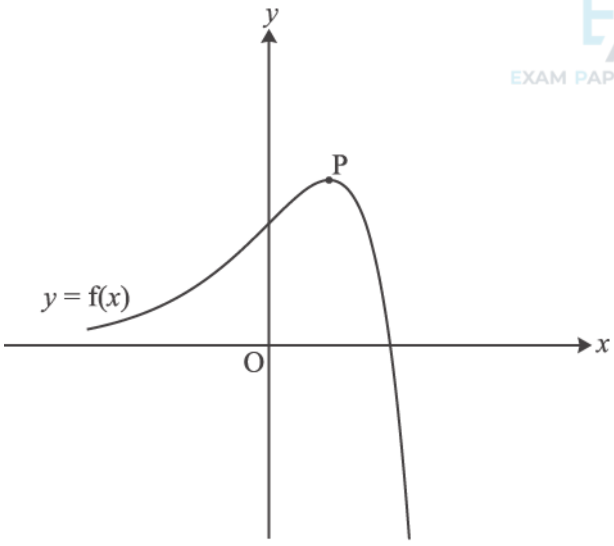


Fig. 8

(i) Write down the coordinates of the intercepts of $y = f(x)$ with the x- and y-axes.

[2]

(ii) Find the exact coordinates of the turning point P.

[6]

(iii) Show that the exact area of the region enclosed by the curve and the x- and y-axes is $\frac{1}{4}\left(e^2 - 3\right)$.

[5]

The function $g(x)$ is defined by $g(x) = 3f\left(\frac{1}{2}x\right)$.

(iv) Express $g(x)$ in terms of x.
Sketch the curve $y = g(x)$ on the copy of Fig. 8, indicating the coordinates of its intercepts with the x- and y-axes and of its turning point.

[4]

(v) Write down the exact area of the region enclosed by the curve $y = g(x)$ and the x- and y-axes.

[1]

18 Fig. 1 shows the graphs of $y = |x|$ and $y = a|x + b|$, where a and b are constants. The intercepts of $y = a|x + b|$ with the x- and y-axes are $(-1, 0)$ and $\left(0, \frac{1}{2}\right)$ respectively.

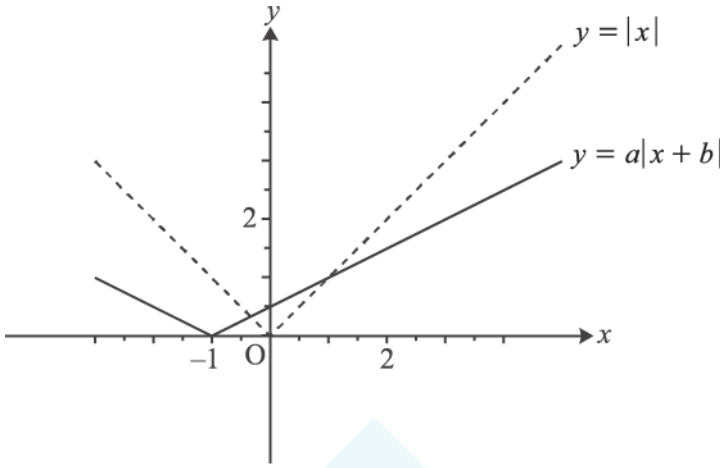


Fig. 1

(i) Find a and b.

[2]

(ii) Find the coordinates of the two points of intersection of the graphs.

[4]

19 Fig. 4 shows the curve $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = a + \cos bx,\ 0 \leqslant x \leqslant 2\pi,$$

and a and b are positive constants. The curve has stationary points at (0, 3) and (2π, 1).

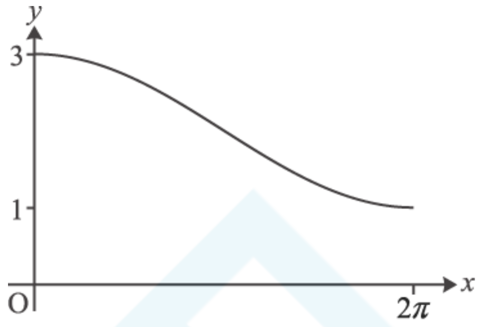


Fig. 4

(i) Find a and b. [2]

(ii) Find $\mathrm{f}^{-1}(x)$, and state its domain and range. [5]

20 Fig. 9 shows the curve $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (\mathrm{e}^x - 2)^2 - 1,\ x \in \mathbb{R}.$$

The curve crosses the x-axis at O and P, and has a turning point at Q.

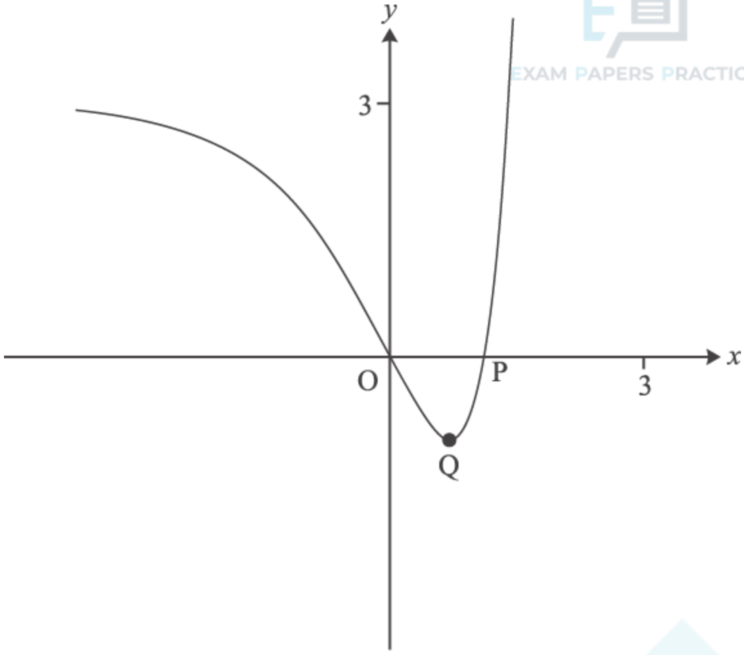


Fig. 9

(i) Find the exact x-coordinate of P.

[2]

(ii) Show that the x-coordinate of Q is ln 2 and find its y-coordinate.

[4]

(iii) Find the exact area of the region enclosed by the curve and the x-axis.

[5]

The domain of $f(x)$ is now restricted to $x \geq \ln 2$.

(iv) Find the inverse function $f^{-1}(x)$. Write down its domain and range, and sketch its graph on the copy of Fig. 9.

[7]

21

Fig. 8 shows the line $y = 1$ and the curve $y = f(x)$, where $f(x) = \frac{(x-2)^2}{x}$. The curve touches the x-axis at P(2, 0) and has another turning point at the point Q.

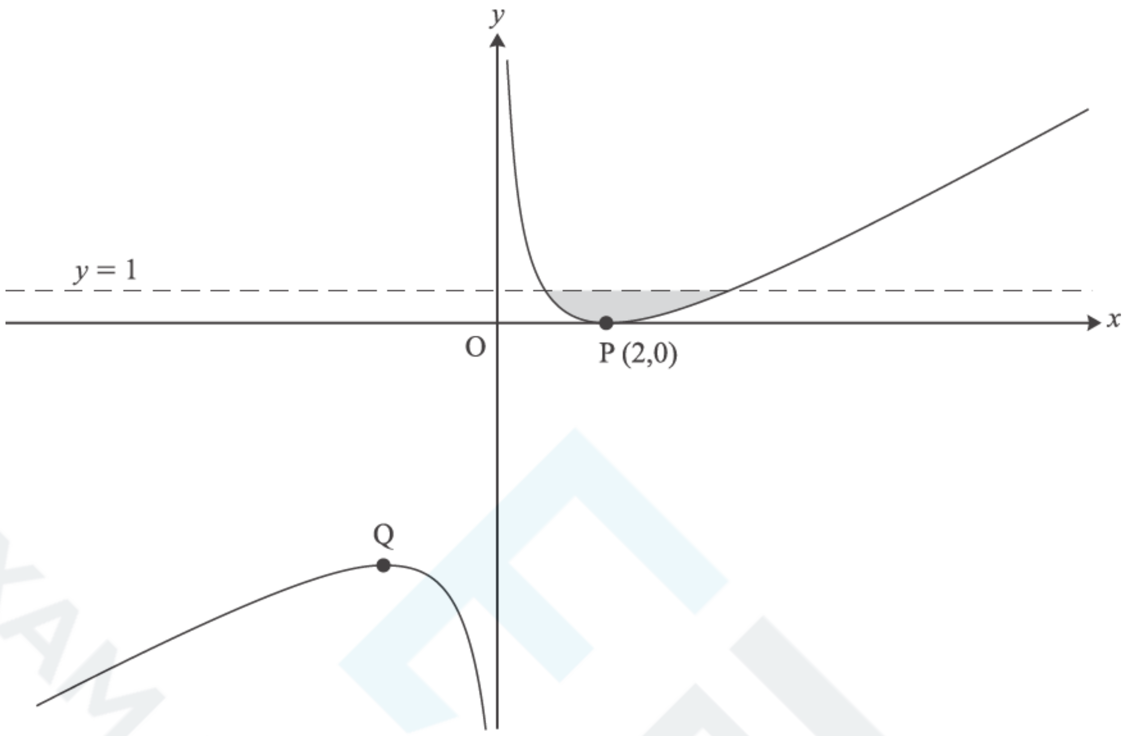


Fig. 8

(i) Show that $f'(x) = 1 - \frac{4}{x^2}$, and find $f''(x)$.

Hence find the coordinates of Q and, using $f''(x)$, verify that it is a maximum point.

[7]

(ii) Verify that the line $y = 1$ meets the curve $y = f(x)$ at the points with x-coordinates 1 and 4. Hence find the exact area of the shaded region enclosed by the line and the curve.

[6]

The curve y = f(x) is now transformed by a translation with vector $\begin{pmatrix}-1\\-1\end{pmatrix}$. The resulting curve has equation $y = g(x)$

(iii) Show that $g(x) = \dfrac{x^2 - 3x}{x+1}$.

[3]

(iv) Without further calculation, write down the value of $\int_0^3 g(x)\,dx$, justifying your answer.

[2]

22 Fig. 9 shows the curve $y = f(x)$, where $f(x) = e^{2x} + k\,e^{-2x}$ and k is a constant greater than 1.

The curve crosses the y-axis at P and has a turning point Q.

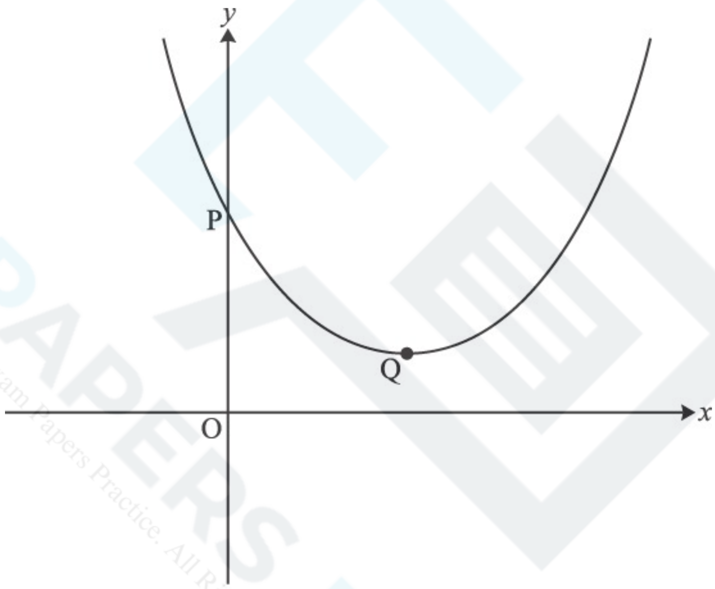


Fig. 9

(i) Find the y-coordinate of P in terms of k.

[1]

(ii) Show that the x-coordinate of Q is $\frac{1}{4}\ln k$, and find the y-coordinate in its simplest form.

[5]

(iii) Find, in terms of k, the area of the region enclosed by the curve, the x-axis, the y-axis and the line $x = \frac{1}{2}\ln k$. Give your answer in the form $ak + b$.

[4]

The function g(x) is defined by $g(x) = f(x + \frac{1}{4}\ln k)$.

(iv)

A Show that $g(x) = \sqrt{k}\,(e^{2x} + e^{-2x})$.

[3]

B Hence show that g(x) is an even function.

[2]

C Deduce, with reasons, a geometrical property of the curve $y = f(x)$.

[3]

23 Fig. 6 shows part of the curve $\sin 2y = x - 1$. P is the point with coordinates $(1.5, \frac{1}{12}\pi)$ on the curve.

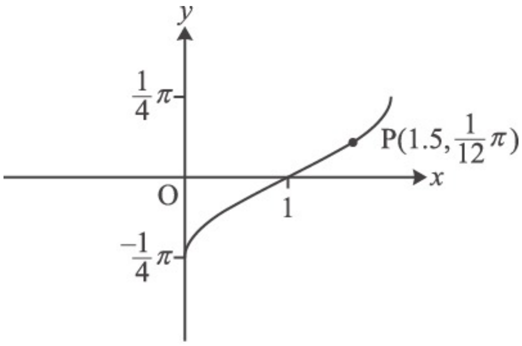


Fig. 6

(i) Find $\frac{dy}{dx}$ in terms of y.

Hence find the exact gradient of the curve $\sin 2y = x - 1$ at the point P.

[4]

The part of the curve shown is the image of the curve $y = \arcsin x$ under a sequence of two geometrical transformations.

(ii) Find y in terms of x for the curve $\sin 2y = x - 1$.

Hence describe fully the sequence of transformations.

[4]

24 By sketching the graphs of $y = |2x + 1|$ and $y = -x$ on the same axes, show that the equation $|2x + 1| = -x$ has two roots. Find these roots.

[4]

25 **(a)** Express $x^2 + 4x + 7$ in the form $(x + b)^2 + c$. **[2]**

(b) Explain why the minimum point on the curve $y = (x + b)^2 + c$ occurs when $x = -b$. **[1]**

26 A circle has equation $(x-2)^2 + (y+3)^2 = 25$.

(a) Write down

- The radius of the circle.
- The coordinates of the centre of the circle. **[2]**

(b) Find, in exact form, the coordinates of the points of intersection of the circle with the y-axis. **[3]**

(c) Show that the point (1, 2) lies outside the circle. **[2]**

(d) The point P(–1, 1) lies on the circle. Find the equation of the tangent to the circle at P. **[4]**

27

(a) Sketch the graph of $y = \frac{1}{x} + a$, where a is a positive constant.

• State the equations of the horizontal and vertical asymptotes.

• Give the coordinates of any points where the graph crosses the axes. **[4]**

(b) Find the equation of the normal to the curve $y = \frac{1}{x} + 2$ at the point where $x = 2$. **[5]**

(c) Find the coordinates of the point where this normal meets the curve again. **[3]**

28 Given that f $(x) = x^3$ and $g(x) = 2x^3 - 1$, describe a sequence of two transformations which maps the curve $y =$ f (x) onto the curve $y = g(x)$. **[4]**

29 In a chemical reaction, the mass m grams of a chemical at time t minutes is modelled by the differential equation

$$\frac{dm}{dt} = \frac{m}{t(1+2t)}.$$

At time 1 minute, the mass of the chemical is 1 gram.

(a) Solve the differential equation to show that $m = \frac{3t}{(1+2t)}$. **[8]**

(b) Hence

(i) find the time when the mass is 1.25 grams, **[2]**

(ii) show what happens to the mass of the chemical as t becomes large. **[2]**

30 (a) The graph of $y = 3\sin^2\theta$ for $0° \le \theta \le 360°$ is shown in Fig. 6. On Fig. 6, sketch the graph of $y = 2\cos\theta$ for $0° \le \theta \le 360°$. [2]

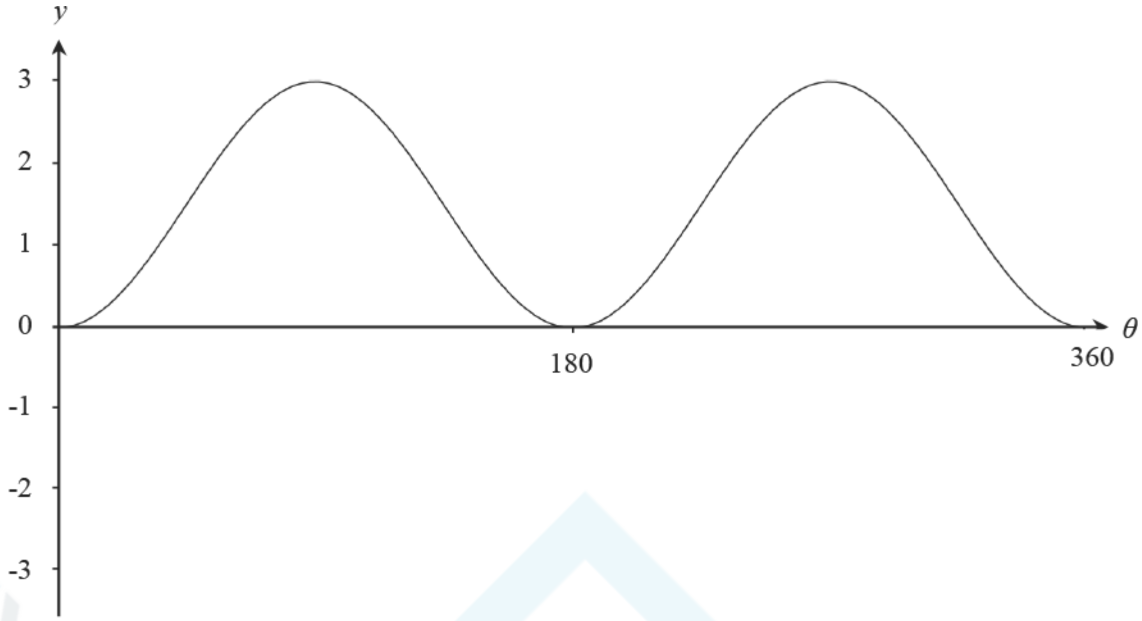


Fig. 6

(b) In this question you must show detailed reasoning.

Determine the values of θ , $0° \leq \theta \leq 360°$, for which the two graphs cross. **[6]**

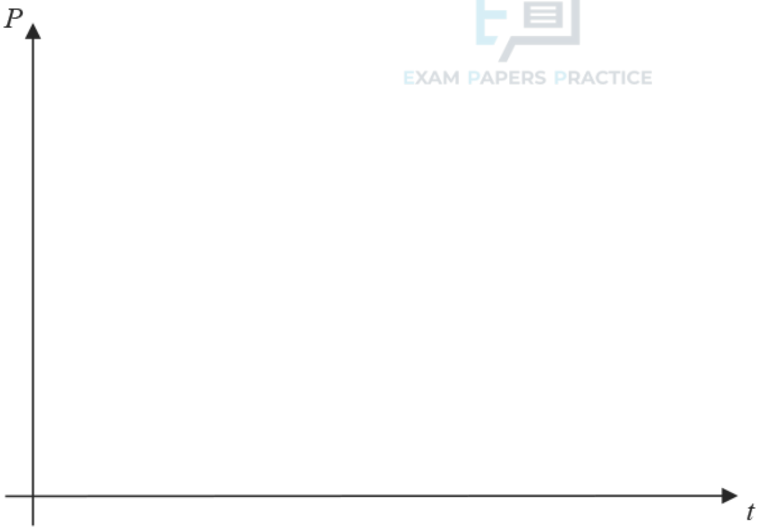
31 In a certain region, the populations of grey squirrels, P_G and red squirrels P_R, at time t years are modelled by the equations:

$$P_G = 10000(1 - e^{-kt})$$

$$P_R = 20000e^{-kt}$$

where $t \geq 0$ and k is a positive constant.

(a) (i) On the axes below, sketch the graphs of P_G and P_R on the same axes.



(ii) Give the equations of any asymptotes. **[4]**

(b) What does the model predict about the long term population of

- grey squirrels
- red squirrels? **[2]**

Grey squirrels and red squirrels compete for food and space. Grey squirrels are larger and more successful than red squirrels.

(c) Comment on the validity of the model given by the equations, giving a reason for your answer. **[1]**

(d) Show that, according to the model, the rate of decrease of the population of red squirrels is always double the rate of increase of the population of grey squirrels. **[4]**

(e) When $t = 3$, the numbers of grey and red squirrels are equal. Find the value of k. **[4]**

32 Fig. 11 shows the curve with parametric equations

$$x = 2\cos\theta,\ y = \sin\theta,\ 0 \le \theta \le 2\pi.$$

The point P has parameter $\frac{1}{4}\pi$. The tangent at P to the curve meets the axes at A and B.

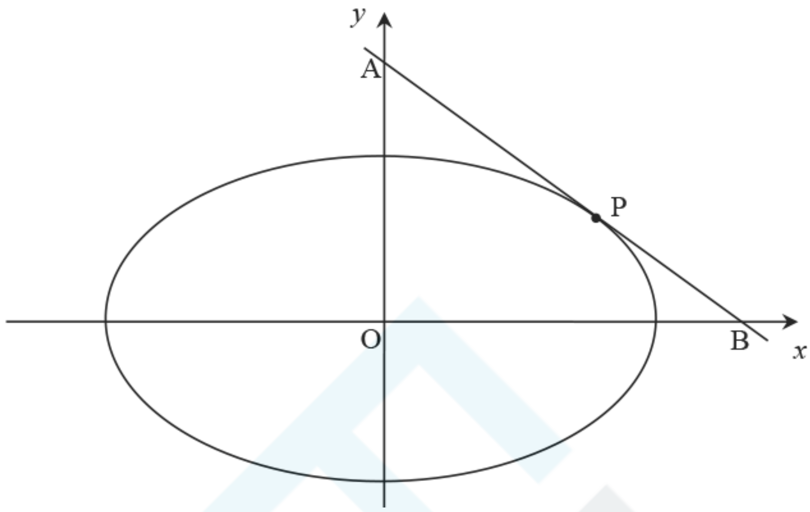


Fig.11

(a) Show that the equation of the line AB is $x + 2y = 2\sqrt{2}$. [6]

(b) Determine the area of the triangle AOB. **[3]**

33 **In this question you must show detailed reasoning.**

Determine the values of k for which part of the graph of $y = x^2 - kx + 2k$ appears below the x-axis. **[4]**

34 **(a)** Sketch the curve $y = e^{2x}$. **[2]**

(b) Describe fully the transformation that maps the curve $y = e^{x}$ onto the curve $y = e^{2x}$. **[2]**

(c) Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 3$, giving your answer in the form $y = e^{a}(bx + c)$ where a, b and c are integers. **[6]**

35 Cheung wishes to model the fuel consumption of a car. He tries the quadratic model

$$y = a(v - b)^2 + c$$

where y is the fuel needed in litres per 100 km and v is the speed in km h^{-1}.

Travelling as a passenger, he notices that the minimum fuel consumption displayed on the dashboard is 10 litres per 100 km and occurs at 80 km h^{-1}.

(a) Write down the values of b and c for which Cheung's model fits this information. **[2]**

At 30 km h^{-1} the fuel consumption displayed is 12 litres per 100 km.

(b) Find the value of a for which Cheung's model fits this information. **[2]**

(c) Use this model to predict the fuel consumption at 110 km h^{-1}. **[2]**

(d) Sketch the graph of the model for speeds between 30 km h^{-1} and 110 km h^{-1}. **[2]**

Later in the journey, Cheung notices that fuel consumption of 12 litres per 100 km is displayed at 30 km h^{-1} and also at 110 km h^{-1}. The minimum fuel consumption still occurs at 80 km h^{-1}.

(e) Give a reason why a quadratic model cannot fit all the information Cheung has found. **[1]**

36 Gerry has been collecting data about the ladybird population in his locality, where they appear mainly during the spring, summer and autumn. He has carried out surveys on many days over a three-month period in the summer, and has recorded the values of N, the number of ladybirds seen, and t, the time in days after 1 January. Using graph-drawing software, he has found that the part of the quadratic curve shown in Fig. 13.1 is a good fit for his data.

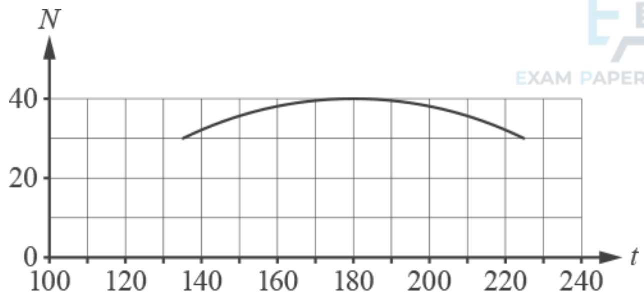


Fig. 13.1

This curve has a maximum point at (180, 40) and the point (135, 30) also lies on the curve. Gerry uses the equation of the curve in the form

$$N = a - b\,(t - c)^{2}$$

as a model for the ladybird population.

(a) Find the values of a and c and show that $b = \frac{2}{405}$. **[3]**

(b) (i) Find the number of ladybirds predicted by the model when $t = 250$. **[1]**

(ii) Explain why this predicted value may be unreliable. **[1]**

(c) (i) Explain why the model is not valid for $t < 90$. **[1]**

(ii) Give another range of values of t for which the model is not valid. **[1]**

With the software set to degrees, Gerry also produces the graph in Fig. 13.2 to model his data for suitable values of t. This graph is a transformation of $N = \cos t$; it passes through (90, 0) and has a maximum point at (180, 40).

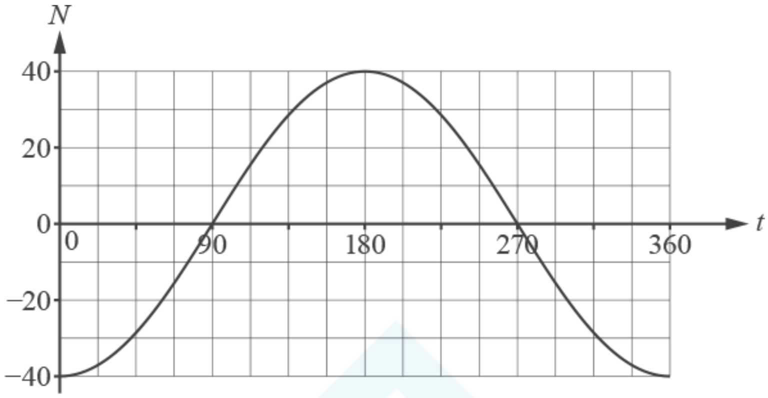


Fig. 13.2

(d) Describe a sequence of transformations that maps the curve $N = \cos t$ to the curve in Fig. 13.2. **[2]**

(e) Write down the equation for N in terms of t for this model. **[1]**

(f) (i) Explain how the model can be refined to have a period of 365 days. **[1]**

(ii) What effect, if any, does this refinement have on the maximum point of the graph? **[1]**

37 **In this question you must show detailed reasoning.**

Fig. 2 shows the line $2x + y = 6$. The line crosses the x-axis at A and the y-axis at B.

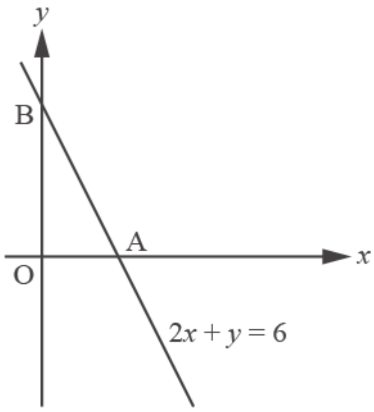


Fig. 2

Find the equation of the quadratic curve which touches the x-axis at A and passes through B. **[5]**

38 (a) In this question you must show detailed reasoning.

Determine the exact values of k for which the curves $y = x^2 - kx$ and $y = 3(k + 1) + kx - x^2$ touch. **[6]**

(b) Determine whether or not there is a value of k for which the curves cross on the y-axis. **[4]**

39 **In this question you must show detailed reasoning.**

(a) For the curve $y = x^4 - 3x^2 + 2$, find the equation of the normal at the point (1, 0). **[4]**

(b) For the curve $y = x^4 - 3x^2 + c$, L is the normal at the point where $x = 1$. Determine the value of c for which L passes through the origin. **[3]**

40 A cubic function $f(x)$ is given by $f(x) = (x - 2)(2x - 3)(x + 5)$.

(i) Sketch the graph of $y = f(x)$. **[3]**

(ii) The curve $y = f(x)$ is translated by $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$. The equation of the translated curve is $y = g(x)$. Show that $g(x) = 2x^3 + 21x^2 + 43x + 24$. **[3]**

(iii) Show that $x = -2$ is one root of the equation $g(x) = 6$ and hence find the other two roots of this equation, expressing your answers in exact form. **[6]**

41 **(i)** Express $y = x^2 + x + 3$ in the form $y = (x + m)^2 + p$ and hence explain why the curve $y = x^2 + x + 3$ does not intersect the x-axis. **[4]**

(ii) Find the coordinates of the points of intersection of the curves $y = x^2 + x + 3$ and $y = 2x^2 - 3x - 9$. **[4]**

(iii) Find the set of values of k for which the curves $y = x^2 + x + k$ and $y = 2x^2 - 3x - 9$ do **not** intersect. **[4]**

42 **(i)** Describe fully the single transformation that maps the curve $y = x^2 + 3$ onto the curve $y = 2x^2 + 6$ **[2]**

(ii) Describe fully the single transformation that maps the curve $y = 2x^2$ onto the curve $y = 2(x - 3)^2$. **[2]**

43 Fig. 3 shows the curve with equation $y = \sqrt{1 + \ln x}$.

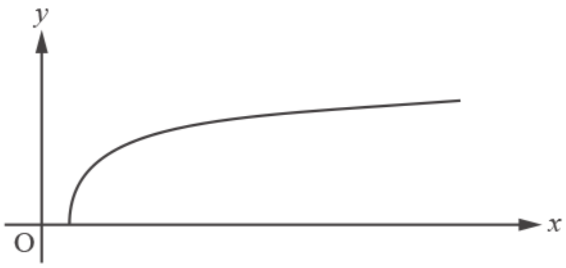


Fig. 3

Determine the exact x-coordinate of the point where the curve meets the x-axis. **[2]**

The equation of a curve is $y = \frac{a}{(x+b)^2}$. Fig. 5 shows the curve for particular values of the constants *a* and *b*.

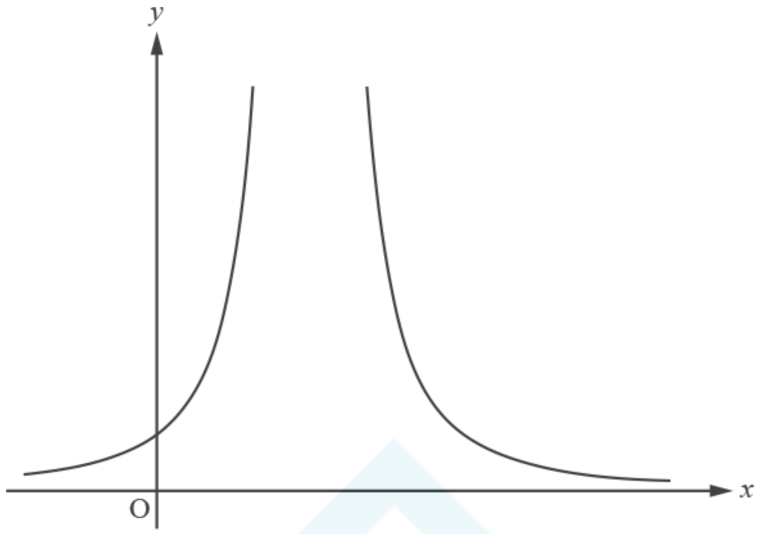


Fig. 5

(a) Write down the equations of the asymptotes, in terms of *a* and/or *b* where necessary. **[2]**

(b) Joe says "For all values of *a* and *b*, the curve lies above the *x*-axis." Determine whether Joe's statement is true or false. **[2]**

(c) The curve goes through the points (1, 3) and (3, 3). Find the values of a and b. **[4]**

45 **In this question you must show detailed reasoning.**

Fig. 8 shows the curve with parametric equations

$$x = 7\cos\theta + 2\cos 2\theta,\ y = 2 + \sin\theta,\ (0 \leq \theta \leq 2\pi).$$

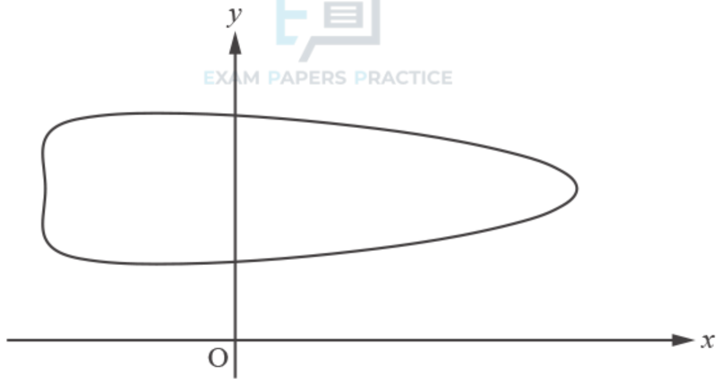


Fig. 8

(a) Find the coordinates of the point on the curve with the greatest y-coordinate. **[4]**

(b) Determine the exact y-coordinates of the points where the curve crosses the y-axis. **[6]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	sketch of cubic the right way up, with two tps	B1		No section to be ruled; no curving back; condone some curving out at ends but not approaching another turning point; condone some doubling (eg erased curves may continue to show); ignore position of turning points for this mark
		i	their graph touching the *x* -axis at –2 and crossing it at 3 and no other places	B1	if intns are not labelled, they must be shown nearby	mark intent if 'daylight' between curve and axis at $x = -2$
		i	intersection of *y*-axis at –12	B1		if no graph but – 12 marked on *y*-axis, or in table, allow this 3rd mark

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	–5 and 0	B2	B1 each; allow B2 for – 5, – 5, 0; or B1 for both correct with one extra value or for (– 5, 0) and (0, 0) or SC1 for both of 1 and 6 **Examiner's Comments** Most candidates obtained full marks for sketching the cubic curve, although their cubics were often unshapely, partly due to the incorrect assumption by many that there was a turning point where the graph crossed the *y*-axis. Most had the cubic the correct way up and realised that it touched the x-axis at –2. A few labelled the y-intersection as 12 rather than –12. A minority sketched parabolas. Full marks were less common in the second part; a small proportion translated to the right rather than to the left as f(x + 3) required. A larger minority did not know what to do and obtained no marks, often giving the single root x = –3.	if their graph wrong, allow –5 and 0 from starting again with eqn, or ft their graph with two intns with *x*-axis
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i	$y = 2x + 3$ drawn accurately	M1	at least as far as intersecting curve twice	ruled straight line and within 2mm of (2, 7) and (–1, 1)
		i	(–1.6 to –1.7, –0.2 to –0.3)	B1	intersections may be in form $x = ..., y = ...$	
		i	(2.1 to 2.2, 7.2 to 7.4)	B1		if marking by parts and you see work relevant to (ii), put a yellow line here and in (ii) to alert you to look
		i	**Revised tolerances for modified papers for visually impaired candidates (graph in (i) with 6mm squares)** $y = 2x + 3$ drawn accurately	M1	at least as far as intersecting curve twice	ruled straight line and within **3** mm of (2, 7) and (–1, 1)
		i	(–1.6 to –1.**8**, –0.2 to –0.3)	B1	intersections may be in form $x = ..., y = ...$	
		i	(2.1 to **2.3**, 7.**1** to 7.4)	B1		if marking by parts and you see work relevant to (ii), put a yellow line here and in (ii) to alert you to look
		i			**Examiner's Comments** Almost all candidates were able to draw the line accurately. Omission of one or both of the signs on the negative intersections was quite common; a few reversed the coordinates. A few just wrote the two x-values only.	
		ii	$\frac{1}{x-2} = 2x + 3$	M1	or attempt at elimination of x by rearrangement and substitution	may be seen in (i) – allow marks; the part (i) work appears at the foot of the image for (ii) so show marks there rather than in (i)
		ii	$1 = (2x + 3)(x - 2)$	M1	condone lack of brackets	implies first M1 if that step not seen

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$1 = 2x^2 - x - 6$ oe	A1	for correct expansion; need not be simplified; NB A0 for $2x^2 - x - 7 = 0$ without expansion seen [given answer]	implies second M1 if that step not seen after $\frac{1}{x-2} = 2x+3$ seen
		ii	$\frac{1 \pm \sqrt{1^2 - 4 \times 2 \times -7}}{2 \times 2}$ oe $\frac{1 \pm \sqrt{57}}{4}$ isw	M1	use of formula or completing square on given equation, with at most one error	completing square attempt must reach at least [2]$(x - a)^2 = b$ or $(2x - c)^2 = d$ stage oe with at most one error
		ii		A1	isw e.g. coordinates; after completing square, accept $\frac{1}{4} \pm \sqrt{\frac{57}{16}}$ or better **Examiner's Comments** Most were able to obtain the correct equation and many went on to solve it successfully, although as expected, there were some errors in using the formula, especially frequently in evaluating the discriminant after correct substitution.	
		iii	$\frac{1}{x-2} = -x + k$ and attempt at rearrangement	M1		
		iii	$x^2 - (k + 2)x + 2k + 1[=0]$	M1	for simplifying and rearranging to zero; condone one error; collection of x terms with bracket not required	e.g. M1 bod for $x^2 - (k + 2)x + 2k$ or M1 for $x^2 - 2kx + 2k + 1[= 0]$
		iii	$b^2 - 4ac = 0$ oe seen or used	M1		= 0 may not be seen, but may be implied by their final values of k

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	[k =] 0 or 4 as final answer, both required	A1	SC1 for 0 and 4 found if 3rd M1 not earned (may or may not have earned first two Ms) **Examiner's Comments** After the previous part, most candidates realised that they had to equate the two expressions and manipulate the resulting equation, although many had problems dealing with the 'k' terms ('$kx + 2x = 2kx$' for instance). Most candidates stopped there, but some realised that they needed to use '$b^2 - 4ac = 0$' to establish the final values of k. Some were confused with the k and x terms and were unable to identify the coefficients correctly or made errors in simplifying the equation. A few candidates used their graphs to establish the results for k. A few tried to apply calculus but rarely with any success.	e.g. obtained graphically or using calculus and / or final answer given as a range
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3			$3(x-2)^2 - 7$ isw or $a = 3$, $b = 2$ $c = 7$ www	M4.	B1 each for $a = 3$, $b = 2$ oe and B2 for $c = 7$ oe or M1 for $[-]\frac{7}{3}$ or for $5 - their\ a(their\ b)^2$ or for $\frac{5}{3} - (their\ b)^2$ soi	condone omission of square symbol; ignore ‘= 0’ may be implied by their answer
			–7 or ft	B1	B0 for (2, –7) **Examiner's Comments** Some who completed the square correctly lost the final mark by giving the minimum point of (2, –7) rather than the minimum *y* -value. Most common part-correct answers were getting the values of *a* and *b* correct but ignoring the multiple of 3 in establishing any value of *c*. The most common wrong values of *b* were –6 (dividing the ‘–12*x*’ by 2) and 4 (taking the 3 out as a common factor and forgetting to divide by 2).	may be obtained by starting again eg with calculus
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	sketch of cubic the right way up, with two tps and clearly crossing the *x* axis in 3 places	B1		no section to be ruled; no curving back; condone slight 'flicking out' at ends but not approaching another turning point; condone some doubling (eg erased curves may continue to show); accept min tp on *y*-axis or in 3^{rd} or 4^{th} quadrant; curve must clearly extend beyond the *x* axis at both 'ends'
		i	crossing / reaching the *x* -axis at –4, –2 and 1.5	B1	intersections must be shown correctly labelled or worked out nearby; mark intent	accept curve crossing axis halfway between 1 and 2 if 3/2 not marked
		i	intersection of *y*-axis at –24	B1		NB to find –24 some are expanding f(*x*) here, which gains M1 in iiiA. If this is done, put a yellow line here and by (iii)A to alert you; this image appears again there
		i			**Examiner's Comments** Most candidates were able to sketch the correct shape for the cubic (the correct way up) and the majority were also able to correctly label the interceptions on the x-axis, although some gave the positive x intercept as ½ or 2/3 or 3. A few candidates failed to label the y-intercept or gave a wrong value such as 12 or -12. Some candidates drew their graph stopping at one of the roots (usually when x = – 4) instead of crossing the x-axis. Only a small number of candidates drew the graph upside-down and a handful drew the wrong shape altogether.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	–2, 0 and 7/2 oe isw or ft their intersections	2	B1 for 2 correct or ft or for (–2, 0) (0, 0) and (3.5, 0) or M1 for $(x + 2)\ x\ (2x - 7)$ oe or SC1 for –6, –4 and –1/2 oe **Examiner's Comments** Quite a few errors were seen here, although a minority knew what to do and wrote down the correct values. Some gave factors or coordinates instead of roots, some solved x – 2 = 0 to give x = 2 as the root, and some went back to the equation but made an algebraic error in replacing x with x – 2, reaching $2x - 5$ as a factor instead of $2x - 7$.	
		iii	*(A)* correct expansion of product of 2 brackets of f(x)	M1	need not be simplified; condone lack of brackets for M1 or allow M1 for expansion of all 3 brackets, showing all terms, with at most one error: $2x^3 + 4x^2 + 8x^2 - 3x^2 + 16x - 12x - 6x - 24$	e.g. $2x^2 + 5x - 12$ or $2x^2 + x - 6$ or $x^2 + 6x + 8$ may be seen in (i) – allow the M1; the part (i) work appears at the foot of the image for (iii)A, so mark this rather than in (i)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	correct expansion of quadratic and linear and completion to given answer	A1	for correct completion if all 3 brackets already expanded, with some reference to show why –24 changes to –9 **Examiner's Comments** The first part was generally well done; most correctly expanded two brackets and continued to simplify and add 15 to get the required result. Common errors were: not dealing correctly with the 15 such as saying g(x) = –15 to get the result, and errors in expanding or collecting terms. There was some poor 'mathematical grammar' with the '+15' often appearing out of nowhere.	condone lack of brackets if they have gone on to expand correctly; condone '+15' appearing at some stage NB answer given; mark the whole process
		iii	*(B)* g(1) = 2 + 9 – 2 – 9 [=0]	B1	allow this mark for $(x – 1)$ shown to be a factor and a statement that this means that $x = 1$ is a root [of $g(x) = 0$] oe	B0 for just $g(1) = 2(1)^3 + 9(1)^2 – 2(1) – 9$ [=0]
		iii	attempt at division by $(x – 1)$ as far as $2x^3 – 2x^2$ in working	M1	or inspection with at least two terms of quadratic factor correct	M0 for division by $x + 1$ after g(1) = 0 unless further working such as g(–1) = 0 shown, but this can go on to gain last M1A1
		iii	correctly obtaining $2x^2 + 11x + 9$	A1	allow B2 for another linear factor found by the factor theorem	NB mixture of methods may be seen in this part – mark equivalently eg three uses of factor theorem, or two uses plus inspection to get last factor;
		iii	factorising a correct quadratic factor	M1	for factors giving two terms correct; eg allow M1 for factorising $2x^2 + 7x – 9$ after division by $x + 1$	allow M1 for $(x + 1)(x + 18/4)$ oe after –1 and –18/4 oe correctly found by formula

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$(2x + 9)(x + 1)(x - 1)$ isw	A1	allow $2(x + 9/2)(x + 1)(x - 1)$ oe; dependent on 2nd M1 only; condone omission of first factor found; ignore '= 0' seen **Examiner's Comments** In part (*B*) most candidates correctly showed g(1) = 0 although some failed to show enough working. Candiates were well-versed, in general, with the techniques of long division or inspection so that most achieved the correct quadratic factor and were able to go on and factorise this to gain full marks. Some tried to use the quadratic formula and then only gave $(x + 1)(x + 4.5)$ oe as factors.	SC alternative method for last 4 marks: allow first M1A1 for $(2x + 9)(x^2 - 1)$ and then second M1A1 for full factorisation
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	$y = (x + 5)(x + 2)(2x – 3)$ or $y = 2(x + 5)(x + 2)(x – 3/2)$	2	M1 for $y = (x + 5)(x + 2)(x – 3/2)$ or $(x + 5)(x + 2)(2x – 3)$ with no equation or $(x + 5)(x + 2)(2x – 3) = 0$ but M0 for $y = (x + 5)(x + 2)(2x – 3) – 30$ or $(x + 5)(x + 2)(2x – 3) = 30$ etc **Examiner's Comments** Candidates struggled with this question. Often they managed to produce the product of binomial factors $x + 2$ $x + 5$ $x – 1.5$ and failed to put it equal to y or put it equal to 0. Those who did have the correct product still very often had an expression only or equated to 0. Many candidates thought that the information about the y-intercept indicated that they should perform a vertical translation and an answer of $y = x + 2$ $x + 5$ $x – 1.5 – 30$ was fairly common among weaker students. Some candidates had an epiphany in part (ii) when they realised that their coefficients should be twice the size and sensibly went back to this part and corrected their error.	allow 'f(x) =' instead of 'y =' ignore further work towards (ii) but do not award marks for (i) in (ii)
		ii	correct expansion of a pair of their linear two-term factors ft isw	M1	ft their factors from (i); need not be simplified; may be seen in a grid	allow only first M1 for expansion if their (i) has an extra –30 etc

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	correct expansion of the correct linear and quadratic factors and completion to given answer $y = 2x^3 + 11x^2 - x - 30$	M1	must be working for this step before given answer or for direct expansion of all three factors, allow M2 for $2x^3 + 10x^2 + 4x^2 - 3x^2 + 20x - 15x - 6x - 30$ oe (M1 if one error) or M1M0 for a correct direct expansion of $(x + 5)(x + 2)(x - 3/2)$ condone lack of brackets if used as if they were there **Examiner's Comments** Many scored only one mark in this part, for correctly expanding a pair of their binomial factors, even after making an error in part (i). As said previously, the light dawned for many in this question and it was good to see that some of these made corrections to part (i). However, many did not and very often there would be a multiplication by 2 done at the end — with or without some attempt at justification for it.	do not award 2nd mark if only had $(x - 3/2)$ in (i) and suddenly doubles RHS at this stage condone omission of 'y =' or inclusion of '= 0' for this second mark (some cands have already lost a mark for that in (i)) allow marks if this work has been done in (i) – mark the copy of (i) that appears below the image for (ii)
		iii	ruled line drawn through (–2, 0) and (0, 10) and long enough to intersect curve at least twice	B1	tolerance half a small square on grid at (–2, 0) and (0, 10)	insert BP on spare copy of graph if not used, to indicate seen – this is included as part of image, so scroll down to see it

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	–5.3 to –5.4 and 1.8 to 1.9	B2	B1 for one correct ignore the solution –2 but allow B1 for both values correct but one extra or for wrong 'coordinate' form such as (1.8, –5.3) **Examiner's Comments** Candidates found this straightforward on the whole, with many scoring full marks for this part. Nearly all drew an accurate line of sufficient length to intersect the curve in three places. Occasionally some read the scale incorrectly when finding the negative solution or were careless with signs, omitting the negative when writing it down.	accept in coordinate form ignoring any y coordinates given
		iv	$2x^3 + 11x^2 - x - 30 = 5x + 10$	M1	for equating curve and line; correct eqns only	annotate this question if partially correct
		iv	$2x^3 + 11x^2 - 6x - 40$ [= 0]	M1	for rearrangement to zero, condoning one error	
		iv	division by $(x + 2)$ and correctly obtaining $2x^2 + 7x - 20$	M1	or showing that $(x + 2)(2x^2 + 7x - 20) = 2x^3 + 11x^2 - 6x - 40$, with supporting working	
		iv	substitution into quadratic formula or for completing the square used as far as $\left(x+\frac{7}{4}\right)^2 = \frac{209}{16}$ oe	M1	condone one error eg a used as 1 not 2, or one error in the formula, using given $2x^2 + 7x - 20 = 0$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$[x =]\dfrac{-7 \pm \sqrt{209}}{4}$ oe isw	A1	dependent only on 4th M1 **Examiner's Comments** There were many attempts to substitute their answers from part (iii) into the given quadratic. Many candidates did not know how to obtain this quadratic, although most eventually went on to attempt to solve it using the formula, sometimes making arithmetic errors in so doing. Of those who did attempt to derive the quadratic, there were several attempts at equating the wrong pair of equations. Some who started correctly expected to see the given answer immediately and stopped at the simplified cubic they had obtained, sometimes having an erroneous –20. Relatively few were able to show that the quadratic factor was the required one, by tong division or by showing multiplying out. A very few candidates used an elegant method of equating the line and cubic and using the factorised form of each to cancel a factor of $x + 2$ on both sides before simplifying.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	$x = 4$	B1		
		i	$(4, -3)$	B1	or $x = 4$, $y = -3$ **Examiner's Comments** The minimum point was generally well found, although some just gave the y coordinate. The question said, "Write down ...", which should suggest to candidates that differentiating and putting the differential equal to zero was not needed. The line of symmetry was also usually well done, but some gave $x = -4$ or $y = 4$.	condone 4, –3
		ii	(0, 13) isw	1	or [when $x = 0$], $y = 13$ isw 0 for just (13, 0) or $(k, 13)$ where $k \neq 0$	annotate this question if partially correct
		ii	[when $y = 0$,] $(x - 4)^2 = 3$	M1	or $x^2 - 8x + 13$ [= 0]	may be implied by correct value(s) for x found allow M1 for $y = x^2 - 8x + 13$ only if they go on to find values for x as if y were 0
		ii	$[x =]4 \pm \sqrt{3}$ or $\frac{8 \pm \sqrt{12}}{2}$ isw	A2	need not go on to give coordinate form A1 for one root correct **Examiner's Comments** The y intercept was usually correct. For the x intercept, many went the long way round: expanding brackets and then using the quadratic formula rather than using the completing the square method.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	replacement of x in their eqn by $(x-2)$	M1	may be simplified; eg [y =] $(x-6)^2-3$ or allow M1 for $(x-6-\sqrt{3})(x-6+\sqrt{3})$ [=0 or y]	condone omission of 'y =' for M1, but must be present in final line for A1
		iii	completion to given answer $y = x^2 - 12x + 33$, showing at least one correct interim step	A1	cao; condone using f$(x-2)$ in place of y **Examiner's Comments** Some candidates lost a mark as they forgot that an equation has 2 sides and omitted the 'y=', only giving an expression. Most candidates realised that they should replace x with $(x-2)$. A minority expanded brackets before replacing x with $(x-2)$ which was a less efficient method.	
		iv	$x^2 - 12x + 33 = 8 - 2x$ or $(x-6)^2 - 3 = 8 - 2x$	M1	for equating curve and line; correct eqns only; or for attempt to subst $(8-y)/2$ for x in $y = x^2 - 12x + 33$	annotate this question if partially correct
		iv	$x^2 - 10x + 25 = 0$	M1	for rearrangement to zero, condoning one error such as omission of '= 0'	
		iv	$(x-5)^2$ [= 0]	A1	or showing $b^2 = 4ac$	allow $\frac{10 \pm \sqrt{0}}{2}$ oe if $b^2 - 4ac$ = 0 is not used explicitly A0 for $(x-5)^2 = y$
		iv	$x = 5$ www [so just one point of contact]	A1	may be part of coordinates $(5, k)$	allow recovery from $(x-5)^2 = y$
		iv	point of contact at $(5, -2)$	A1	dependent on previous A1 earned; allow for $y = -2$ found	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv				examiners: use one mark scheme or the other, to the benefit of the candidate if both methods attempted, but do not use a mixture of the schemes
		iv	for curve, $y' = 2x - 12$	M1		
		iv	$2x - 12 = -2$	M1	for equating their y' to -2	
		iv	$x = 5$, and y shown to be -2 using eqn to curve	A1		
		iv	tgt is $y + 2 = -2(x - 5)$	A1		
		iv	deriving $y = 8 - 2x$	A1	**Examiner's Comments** Since both equations were given, those were the ones which had to be used, and most candidates did so successfully. Some candidates did not realise that obtaining $(x - 5)^2 = 0$ led to sufficient evidence of a repeated root and also showed that the discriminant was zero. Some, of course, did not attempt to factorise anyway but opted for using the formula. The main error was in the very last mark, where some candidates substituted their x value back into the quadratic that they had just solved to find $y = 0$, rather than using the line or the curve to give $y = -2$.	condone no further interim step if all working in this part is correct so far
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	(0, –3)	B1	condone $y = -3$, isw	if not coordinates, must be clear which is x and which is y
		i	(– 1/2 , 0) and (3, 0) www	B2	condone $y = -1/2$, and 3; **B1** for one correct www or **M1** for $(2x + 1)(x - 3)$ or correct use of formula or reversed coordinates	**Examiner's Comments** Many candidates earned all 3 marks in this part. Some forgot to find the y-intercept. A few used the quadratic formula or completed the square, perhaps not realising that factorising was possible.
		ii	$2x^2 - 6x - 6[= 0]$ isw or $x^2 - 3x - 3$ [= 0] or $2y^2 - 18y + 30$ [=0]	M1	for equating curve and line, and rearrangement to zero, condoning one error	allow rearranging to constant if they go on to attempt completing the square
		ii	use of formula or completing the square, with at most one error	M1	no ft from $2x^2 - 6x = 0$ or other factorisable equations	if completing the square must get to the stage of complete square only on lhs as in 9(ii)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\left(\frac{6\pm\sqrt{84}}{4},\frac{18\pm\sqrt{84}}{4}\right)$ or $\left(\frac{3\pm\sqrt{21}}{2},\frac{9\pm\sqrt{21}}{2}\right)$ oe isw	A2	**A1** for one set of coords or for x values correct (or ys from quadratic in y); need not be written as coordinates	A0 for unsimplified y coords eg $\frac{3+\sqrt{21}}{2}+3$ **Examiner's Comments** The majority of candidates chose the straightforward approach and equated the line and curve given. These candidates most often simplified correctly to a required form and applied the formula. This was often very well carried out. A very good number of candidates earned 3 marks using this approach. Some attempted to complete the square. Those who, sensibly, divided through by 2 before doing so were usually successful – those who did not were less successful. Most candidates struggled to find and simplify the y -coordinates. Some simply omitted them and the many who attempted them often just wrote the x coordinates '+ 3' or failed to convert the 3 being added to a fraction of a common denominator to add to the x-coordinate. Some candidates made their solution unnecessarily complicated by rearranging the equation of the line and substituting for x. These candidates often omitted to take the solitary y-term into account and mostly scored no more than the first two marks.
		iii	$2x^2 - 5x - 3 = x + k$	M1	for equating curve and line	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$2x^2 - 6x - 3 - k$ [= 0]	M1	for rearrangement to zero, condoning one error, but must include k ;this second M1 implies the first, eg it may be obtained by subtracting the given equations	
		iii	$b^2 - 4ac < 0$ oe for non-intersecting lines	M1	eg allow for just quoting this condition; may be earned near end with correct inequality sign used there allow 'discriminant is negative' if further work implies $b^2 - 4ac$	some may use condition for intersecting lines or for a tangent and then swap condition at the end; only award this M1 and the final A mark if the work is completely clear
		iii	$36 - 8 \times -(3 + k)$ [< 0] oe	A1	for correct substitution into $b^2 - 4ac$; no ft from wrong equation; if brackets missing or misplaced, must be followed by a correct simplified version	can be earned with equality or wrong inequality, or in formula – this mark is not dependent on the 3rd M mark;
		iii	$k < -\frac{15}{2}$ oe		isw if 3rd M1 not earned, allow **B1** for $-\frac{15}{2}$ obtained for k with any symbol	
		iii	or, for those using a tangent condition with trials to find the boundary value			mark one mark scheme or another, to the advantage of the candidate, but not a mixture of schemes
		iii	rearrangement with correct boundary value of k eg $2x^2 - 6x + 4.5$ [= 0] or $2x^2 - 6x - (3 - 7.5)$ [= 0]	M2	**M1** for $2x^2 - 5x - 3 = x - 7.5$	M0 for trials with wrong values without further progress, though may still earn an M1 for $b^2 - 4ac < 0$
		iii	showing $36 - 8 \times -(3 - 7.5) = 0$ or $36 - 8 \times 4.5 = 0$ oe	M1	may be in formula implies previous M2	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$k < -\frac{15}{2}$ oe	A2	**B1** for $-\frac{15}{2}$ obtained for k as final answer with any symbol	
		iii	or, for using tangent with differentiation			
		iii	$y = 4x - 5$	M1		
		iii	[when $y = x + k$ is tgt] $4x - 5 = 1$	M1		
		iii	$x = 1.5$, $y = -6$	A1		
		iii	$-6 = 1.5 + k$ or $k = -7.5$ oe	A1		
		iii	$k < -7.5$ oe	A1		**Examiner's Comments** Some candidates were unable to cope with the constant of the equation they had formed being in terms of k. Many equated the line and curve, as before, and found $2x^2 - 6x - 3 - k = 0$ and then, rather than applying $b^2 - 4ac < 0$, they wrote $2x^2 - 6x - 3 = k$ and tried to apply $b^2 - 4ac < 0$ to the left hand side. Those who did work with $2x^2 - 6x - 3 - k = 0$ were almost always successful. Some candidates made sign errors through carelessness. Some introduced wrong brackets into their equation in an attempt to group the c term, such as $-(3 - k)$. Some candidates correctly substituted into $b^2 - 4ac < 0$ but were unable to multiply out correctly. The result $36 - 8 - (3 + k)$ was not uncommon amongst these candidates. Other

Question			Answer/Indicative content	Marks	Part marks and guidance	
						candidates used trials on $2x^2 - 6x - 3 - k = 0$ to find the boundary value ie the constant that gave $b^2 - 4ac = 0$. These often scored 3 marks, but sign errors usually resulted in the loss of the final 2 marks. A very few candidates used a calculus approach. In most cases, once $y' = 4x - 5$ had been found, it was equated to 0 and the minimum point established, thinking that this would be helpful, then no further progress was made. Candidates' setting out in this question was often poor and difficult to make sense of – particularly if they had had several attempts or had used trials. Some candidates lost marks as they restarted several times, with each time being worth less than the previous attempt! Candidates should take care in this regard – and indicate which of their attempts they intend to be taken as the answer in these cases.
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	–5.7 to –5.8, –2.2 to –2.3, –1 isw	2	**B1** for 2 correct or for all 3 only stated in coordinate form, ignoring *y* coordinates	**Examiner's Comments** About the same number of candidates gave the coordinates of intersection of the two graphs as gave the requested roots of the given equation in *x*. A few misread from the graph and/or struggled with the scale.
		ii	$1=(x + 2)(x^2 + 7x + 7)$	M1	condone missing brackets if expanded correctly; or **M1** for correct expansion of $(x + 2)(x^2 + 7x + 7)$	
		ii	correct completion with at least one interim stage of working to given answer: $x^3 + 9x^2 + 21x + 13 = 0$	A1		
		ii	[$x = –1$ is root so] $(x + 1)$ is factor soi	M1	implied by division of cubic by $x + 1$	condone some confusion of root/factor for this mark if division of cubic by $x + 1$ seen
		ii	correctly finding other factor as $x^2 + 8x + 13$	M2	**M1** for correct division of cubic by $(x + 1)$ as far as obtaining $x^2 + 8x$ (may be in grid) or for two correct terms of $x^2 + 8x + 13$ obtained by inspection	allow seen in grid without + signs
		ii	$\dfrac{-8 \pm \sqrt{8^2 - 4\times 13}}{2}$ oe	M1	for use of formula, condoning one error, for $x^2 + 8x + 13= 0$	or **M1** for $(x + 4)^2 = 4^2 – 13$ oe or further stage, condoning one error

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{-8 \pm \sqrt{12}}{2}$ isw or $-4 \pm \sqrt{3}$ isw and $x = -1$	A1	x = –1 may be stated earlier	**Examiner's Comments** isw wrong simplification or giving as coordinates **Examiner's Comments** In the main this part was completed well, with almost all candidates gaining the first two marks for multiplying by (x+2) and expanding to prove the stated equality. A significant number of candidates were unable to progress further, unsure of how to solve a cubic equation. Stronger candidates produced a well-organised solution, leading directly to the fully factorised expression (sometimes in only a few lines of working, having used the root of x =–1 from the graph and/or part (i) to obtain the first factor). The majority were able to find the correct quadratic following division by (x + 1), with a few using synthetic division and a sizeable minority finding the solution by inspection. At this stage many found the correct final solution, but a significant number failed to include x =–1 in their final solution to this question, or stated incorrectly that (x + 1) was a root.
		iii	(A) drawing the translated quadratic	B1	must be a reasonable translation of given quadratic, only intersecting given curve once; intersections with x axis –3 to –2.5 and 1.5 to 2; ignore above y = 1	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	or showing that the horizontal gap between the relevant parts of the curve is always less than 3			
		iii	estimated coordinates of the point of intersection (1.8 to 2, 0.2 to 0.3)	B1		**Examiner's Comments** Many candidates were able to translate correctly although there were issues with the intersections on the *x*-axis for some. Quite a few candidates pointed out the intersection but did not write down the coordinates as requested.
		iii	$y = x^2 + x - 5$ or $y = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4}$ (B)	2	**M1** for [*y* =] $(x - 3)^2 + 7(x - 3) + 7$ oe or for simplified equation with '*y* =' omitted or for $y = (x - a)(x - b)$ where *a* and *b* are the values $3 + \frac{-7 \pm \sqrt{21}}{2}$ oe (may have been wrongly simplified)	M0 for use of estimated roots in (*A*) **Examiner's Comments** Many candidates failed to recognise that they should substitute $(x - 3)$ for *x* in the original equation, with a variety of different methods attempted. Substituting $(x + 3)$ was quite a common error, as was adding 3 to the original equation, or changing the constant term to –5. Some used estimated roots. Many failed to gain full marks because they omitted '*y* =' from their final answer.
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	[x =] 5, [x =] –1 www	2	**M1** for x – 2 = ±3 or for (x – 5)(x + 1) [=0]	0 for just x = 5 or for x – 2 = 3
		ii	parabola shape curve the correct way up	1	must extend beyond x-axis;	condone 'U' shape or very slight curving back in/out; condone some doubling / feathering–deleted work sometimes still shows up in rm assessor; must not be ruled; condone fairly straight with clear attempt at curve at minimum; be reasonably generous on attempt at symmetry e.g. condone minimum on y-axis for this mark
		ii	intersecting x-axis at 5 and –1 or ft from (i) and y-axis at –5	1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	turning point (2, –9)	1	seen on graph or identified as tp elsewhere in this part	may be implied by 2 and –9 marked on axes 'opposite' turning point **Examiner's Comments** Most candidates managed to solve the equation. Quite a number of candidates multiplied out the brackets and rearranged to form a quadratic equation in the traditional form. This was then usually solved by factorisation, but occasionally using the formula. Those who used the given form and took the square root of both sides were more inclined to find just one root, by ignoring the possibility that the square root of 9 could be –3. The quality of the parabolas varied enormously, but most candidates determined the coordinates of the turning point and made a good attempt. Some candidates did not consider the turning point and often had skewed parabolas with a minimum on the *y*-axis. A few candidates sketched cubics and received no marks.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	(0.8, –2) oe	2	**B1** each coordinate **Examiner's Comments** The majority obtained both marks. The usual errors were present: (20, –2), (4, -10) and (0.8, -0.4) being the most common, but (9, –2) and (0, –2) were also seen occasionally.	**SC0** for (4, –2)
		ii	Translation	B1		
		ii	$\begin{pmatrix} 90 \\ 0 \end{pmatrix}$ oe	B1	or eg 270 to left **Examiner's Comments** Surprisingly few candidates used the word "translate", and opted for their own terminology such as "move to the right" or "shift to the right". Many candidates identified 90 to the right or gave the appropriate vector form. A few gave ambiguous answers or gave the answer "90 to the left".	allow **B2** for rotation through 180° about (45, 0) oe
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		i	graph from (–1, 1) to (1, 1) to (2, 2) to (3, 0)	2	**B1** for three points correct or for all four points correct but clearly not joined <u>Examiner's Comments</u> The majority of candidates scored full marks. A few lost a mark by extending their function to the left or the right or by misplacing (2, 2) or (3, 0). Approximately 30% of candidates failed to score. A translation of $\begin{pmatrix}-2\\0\end{pmatrix}$ or a stretch in the *x*-direction scale factor 2 were the most common errors; a few candidates gave the end point as (2, 0) and the adjacent vertex as (1.5, 2).	points must be joined, but not always easy to see, so BOD if in doubt. Accept freehand drawing.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	graph from (–2, 3) to (2, 3) to (4, 6) to (6, 0)	2	**B1** for three points correct or for all four points correct but clearly not joined <u>Examiner's Comments</u> There was an even better response to this part with almost 70% of candidates obtaining full marks. As in part (i), a few lost a mark by extending their function to the left or the right or by misplacing (4, 6) or (more often) (6, 0). Approximately one quarter of candidates failed to score. A translation of $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$ or a stretch in the *x*-direction scale factor 3 were common; occasionally (4, 6) and (6, 0) were correct, but the other two points were simply left unaltered.	points must be joined, but not always easy to see, so BOD if in doubt. Accept freehand drawing.
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	(6, –1.5) oe	B2	B1 for each value; Allow $x = 6$, $y = -1.5$ **Examiner's Comments** Both parts were generally very well done, with many candidates scoring full marks. The most common errors were (6, – 6) and (12, – 3) in part (i) and (18, – 3) and (6, – 1)	SC0 for (6, – 3)
		ii	(2, – 3)	B2	B1 for each value; Allow $x = 2$, $y = -3$ **Examiner's Comments** Some candidates applied the scale factor to both values.	SCO for (6, – 3)
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$\left[\frac{dy}{dx}=\right] 4 \times 2 + 3$ or 11 isw	M1*		
		i	9 = their (4 × 2 + 3) × 2 + c	M1dep*	or y – 9 = their (4 × 2 + 3) × (x – 2)	
		i	y = 11x – 13 or y = 11x + c and c = – 13 stated isw	A1	or y – 9 = 11 (x – 2) isw **Examiner's Comments** The majority of candidates gained full marks on this question. A few found the gradient correctly and then went on to find the equation of the normal, and some candidates integrated and found the equation of the curve.	
		ii	$\frac{4x^2}{2}+3x$	M1*		
		ii	[y =] 2x^2 + 3x + c	A1	must see “2” and “+ c”, may be earned later eg after attempt to find c	
		ii	9 = 2 × 22 + 3 × 2 + c	M1dep*	must include constant, which may be implied by answer	
		ii	y = 2x^2 + 3x – 5 cao	A1	allow first 4 marks for y = 2x^2 + 3x + c and c = –5 stated	
		ii	(1, 0) *and* (–2.5, 0) oe cao	B1	or for x = 1, y = 0 and x = –2.5, y = 0	**B0** for just stating x = 1 and x = –2.5
		ii	$x=-\frac{3}{4}$	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y=-\frac{49}{8}$	B1	–6.125 or – 6⅛ Examiner's Comments Most candidates integrated successfully. A few omitted '+ c' and made little progress thereafter, but the majority successfully obtained the equation of the curve. Many candidates failed to give the coordinates in a correct form or transposed the signs, thus losing an accuracy mark. Most candidates used the given derivative to find the co-ordinates of the minimum point, but it was surprising how many made a sign error and then couldn't obtain the correct value for y. A significant number ofcandidates completed the square instead of using the derivative, and most lost accuracy.	
		iii	substitution to obtain [y =] f($2x$) in polynomial form	M1	f (x) must be the quadratic in x with linear and constant term obtained in part (ii), may be in factorised form	or their x = 1 → their 0.5 and their x = – 2.5 → their x = –1.25
		iii	y = ($2x$ – 1)($4x$ + 5) or y = or $y=2\left(2x+\frac{3}{4}\right)^2-\frac{49}{8}$	A1FT	must be simplified to one of these forms, **FT** their quadratic in$x$ with linear and constant term obtained in part (ii)	hence y = ($2x$ – 1)($4x$ + 5) FT their x-intercepts from their quadratic in x with linear and constant term obtained in part (ii)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\left(-\frac{3}{8},-\frac{49}{8}\right)$ oe	B1	or **FT** their (both non-zero) co-ordinates for minimum point or their quadratic in x with linear and constant term obtained in part (ii) **Examiner's Comments** Very few candidates realised that they needed to work with f(2x) to find the newequation. The majority of those who did adopt the correct approach often went wrong, usually with the first term. In order to earn the method mark by this approach examiners needed to see the substitution: many candidates just wrote down $y = 4x^2 + 6x – 5$ and failed to score. It may have been the case that the correct approach was being attempted, but this answer was also seen resulting from totally wrong working! A few candidates successfully worked with the images of the intercepts following the stretch, but often failed to simplify the answer correctly. A minority of candidates earned the third mark, either as a follow through mark or for a fully correct answer. However, many candidates multiplied the x value by 2, or halved the y value instead or as well.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	both curves with positive gradients in 1st and 2nd quadrants; ignore labels for this mark	M1	do not award if clearly not exponential shape; condone touching negative x-axis but not crossing it	consider each curve independently; ignore scales and points apart from (0, 1)
		i	both through (0, 1)	A1		allow if indicated in table of values or commentary if notmarked on graph
		i	$y = 3^{2x}$ above $y = 3^x$ in first quadrant and below it in second	A1	must be clearly labelled, **A0** if wrongly attributed or if coincide for negative x from (0, 1) **Examiner's Comments** A small number of candidates drew two curves of totally different shapes, which was surprising, but most knew the correct shape and although many sketches were sloppily presented, and marks were lost through omitting to identify (0, 1) or by allowing the curves to coalesce through the second quadrant.	if **M0** allow **SC1** for one graph fully correct
		ii	$x = 3$	B1	**B0** if wrongly attributed	
		ii	$3^x = 27$	B1	**B0** if wrongly attributed **Examiner's Comments** This was very well done. Nearly all candidates correctly found $x = 3$; a few thenevaluated 3^3 as 6, 9 or 81.	allow $3^3 = 27$ with $x = 3$ stated
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	sin *kx*	M1	$k > 0$ and $k \neq 1$	condone use of other variable
		i	$y = \sin 2x$	A1	must see "y =" at some stage for **A1** **Examiner's Comments** One or two easy marks were lost in a surprising variety of ways. Many candidates gave the answer as $y = \sin x$, $y = 2\sin x$ or $y = \sin\frac{1}{2}x$ and some omitted "y =".	condone f(x) = sin 2x
		ii	sketch of sine curve with period 360° and amplitude 1	B1	for $0 \leq x \leq 450$; ignore curve outside this range;	amplitude, period and centring on $y = -3$ must be clear from correct numerical scale, numerical labelling
		ii			do not allow sketch of $y = \cos x$ or $y = -\sin x$ for either mark	or comment; strokes on axes insufficient to imply scale: mark intent
		ii	sine curve centred on $y = -3$ and starting at (0, –3)	B1	**Examiner's Comments** Only a few candidates presented good sketches with the key points clearly identified. Too much was often left to the imagination of the marker. Candidates are reminded of the need to indicate amplitude, period and centring by clear scales and labelling. Unnumbered strokes on the axes, for instance, are insufficient. A variety of misunderstandings was evident. $y = \sin(x - 3)$ was a common error, and occasionally $y = 3\sin x$ or $y = -\sin x$ were seen.	allow full marks if $y = \sin x$ and $y = \sin x - 3$ seen on same diagram
			Total	4		

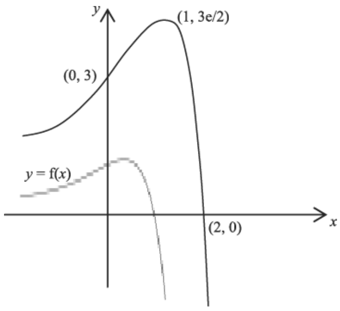
Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	translation in the *x*-direction	M1	allow 'shift', 'move'	If just vectors given withhold **one** 'A' mark only
		i	of π/4 to the right	A1	oe (eg using vector)	'Translate $\begin{pmatrix} \pi/4 \\ 1 \end{pmatrix}$' is 4 marks; if this is followed by an additional incorrect transformation, SC M1M1A1A0
		i	translation in *y*–direction	M1	allow 'shift', 'move'	$\begin{pmatrix} \pi/4 \\ 1 \end{pmatrix}$ only is M2A1A0
		i	of 1 unit up.	A1	oe (eg using vector) **Examiner's Comments** We usually insist on the word 'translation' here, but in this case allowed 'move', 'shift', etc. A vector on its own does not in our view imply a translation. Occasionally, candidates clearly knew what the transformations were, but wrote the vectors incorrectly, for example the wrong way up. Nevertheless, this topic is usually well known and done well.	
		ii	$g(x) = \dfrac{2\sin x}{\sin x + \cos x}$			(Can deal with num and denom separately)
		ii	$g'(x) = \dfrac{(\sin x + \cos x)2\cos x - 2\sin x(\cos x - \sin x)}{(\sin x + \cos x)^2}$	M1	Quotient (or product) rule consistent with their derivs	$\dfrac{vu' - uv'}{v^2}$; allow one slip, missing brackets
		ii	$= \dfrac{2\sin x\cos x + 2\cos^2 x - 2\sin x\cos x + 2\sin^2 x}{(\sin x + \cos x)^2}$ $= \dfrac{2\cos^2 x + 2\sin^2 x}{(\sin x + \cos x)^2} = \dfrac{2(\cos^2 x + \sin^2 x)}{(\sin x + \cos x)^2}$	A1	Correct expanded expression (could leave the '2' as a factor)	$\dfrac{uv' - vu'}{v^2}$
		ii	$= \dfrac{2}{(\sin x + \cos x)^2}$ *	A1	**NB AG**	must take out 2 as a factor or state $\sin^2 x + \cos^2 x = 1$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	When $x = \pi/4$, $g'(\pi/4) = 2/(1/\sqrt{2} + 1/\sqrt{2})^2$	M1	substituting π/4 into correct deriv	
		ii	$= 1$	A1		
		ii	$f'(x) = \sec^2 x$	M1	o.e., e.g. $1/\cos^2 x$	
		ii	$f'(0) = \sec^2(0) = 1$, [so gradient the same here]	A1	**Examiner's Comments** The quotient rule is generally well known, and errors here usually stemmed from faulty derivatives or poor algebra. Brackets are not optional in an expression like this, and their removal was not always successfully achieved. We also needed evidence of the use of $\cos^2 x + \sin^2 x = 1$, either by its direct quotation or by factoring out the '2' in the numerator. The evaluation of $g'(x)$ was usually correct. With $f'(x)$, some used a quotient rule on $\sin x/\cos x$ rather than quoting the derivative of $\tan x = \sec^2 x$; we also got some occasional 'translation' arguments here which misunderstood the nature of the verification.	
		iii	$= \int_1^{1/\sqrt{2}} -\frac{1}{u}\,du$ let $u = \cos x$, $du = -\sin x\,dx$ when $x = 0$, $u = 1$, when $x = \pi/4$, $u = 1/\sqrt{2}$			
		iii	$= \int_{1/\sqrt{2}}^{1} \frac{1}{u}\,du$ *	M1	substituting to get $\int -1/u\ (du)$	ignore limits here, condone no du but not dx allow $\int 1/u.-du$
		iii	$= \int_{1/\sqrt{2}}^{1} \frac{1}{u}\,du$ *	A1	NB **AG**	but for A1 must deal correctly with the -ve sign by interchanging limits

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= [\ln u]_{1/\sqrt{2}}^{1}$	M1	[ln *u*]	
		iii	= ln 1 - ln (1/√2)			
		iii	= ln √2 = ln $2^{½}$ ln 2	A1	ln √2, ½ ln 2 or –ln(1/√2) **Examiner's Comments** This was a case where giving the transformed integral proved to be of doubtful value, as many candidates 'lost' the negative sign in their ∫-1/*u* d*u*, and placed the limits the wrong way round. It appears that the idea of swapping limits making the integral negative was not generally understood. The evaluation of the given integral with respect to *u* was more successfully done, though quite a few candidates approximated their final answer.	mark final answer
		iv	Area = area in part **(iii)** translated up 1 unit.	M1	soi from π/4 added	or
		iv	So = ½ ln 2 + 1 × π4 = ½ ln 2 + π/4.	A1cao	oe (as above) **Examiner's Comments** These marks were gained by candidates who managed to spot the rectangle of area added by the translation upwards of the graph of f(*x*).	
			Total	17		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	(1, 0) and (0, 1)	B1B1	$x = 0$, $y = 1$; $y = 0$, $x = 1$ **Examiner's Comments** The points of intersection were a write-down for many candidates. Weaker attempts failed to solve $(1 - x)\, e^{2x} = 0$ convincingly.	
		ii	$f'(x) = 2(1 - x)e^{2x} - e^{2x}$	B1	$d/dx(e^{2x}) = 2e^{2x}$	
		ii		M1	product rule consistent with their derivatives	
		ii	$= e^{2x}(1 - 2x)$	A1	correct expression, so $(1 - x)e^{2x} - e^{2x}$ is B0M1A0	
		ii	$f'(x) = 0$ when $x = ½$	M1dep	setting their derivative to 0 dep 1st M1	
		ii		A1cao	$x = ½$	
		ii	$y = ½\, e$	B1	allow $½\, e^1$ isw **Examiner's Comments** This proved to be an accessible 6 marks for candidates. The derivative of e^{2x} and the product rule were generally correct, and deriving $x = ½$ and $y = e^{1/2}$ was straightforward, though many did not simplify the derivative to $e^{2x} - 2xe^{2x}$ immediately. Some candidates approximated for $e^{1/2}$ and lost a mark.	
		iii	$A = \int_0^1 (1 - x)e^{2x}\,dx$	B1	correct integral and limits; condone no dx (limits may be seen later)	
		iii	$u = (1 - x),\ u' = -1,\ v' = e^{2x},\ v = ½\, e^{2x}$	M1	u, u', v', v, all correct; or if split up $u = x$, $u' = 1$, $v' = e^{2x}$, $v = ½\, e^{2x}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow \quad A=\left[\frac{1}{2}(1-x)e^{2x}\right]_0^1-\int_0^1 \frac{1}{2}e^{2x}.(-1)dx$	A1	condone incorrect limits; or, from above, $\ldots\left[\frac{1}{2}xe^{2x}\right]_0^1-\int_0^1 \frac{1}{2}e^{2x}dx$	
		iii	$=\left[\frac{1}{2}(1-x)e^{2x}+\frac{1}{4}e^{2x}\right]_0^1$	A1	o.e. if integral split up; condone incorrect limits	
		iii	$= ¼\, e^2 – ½ – ¼$ $= ¼\,(e^2 – 3)$ *	A1cao	NB AG <u>Examiner's Comments</u> Most candidates applied integration by parts to either $\int(1 – x)\, e^{2x}dx$ or $\int x\, e^{2x}dx$, using appropriate u, v', u' and v. Sign and/or bracket errors sometimes meant they failed to derive the correct result, but many were fully correct.	
		iv	g(x) = 3f(½ x) = 3(1 – ½ x) e^x	B1	o.e; mark final answer	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		B1 B1dep B1	through (2,0) and (0,3) – condone errors in writing coordinates (e.g. (0,2)). reasonable shape, dep previous B1 TP at (1, 3e/2) or (1, 4.1) (or better). (Must be evidence that x = 1, y = 4.1 is indeed the TP – appearing in a table of values is not enough on its own.) Examiner's Comments This part proved to be quite demanding. Deriving the formula for g(x) was rarely correctly done. Common errors were an extra factor of 3 and an incorrect exponent. Most graphs showed the correct points of intersection (0, 3) and (2, 0), but the turning point was quite often incorrect or missing, and the shape failed to convince.	
		v	$6 \times \frac{1}{4}(e^2 - 3)$ [$= 3(e^2 - 3)/2$]	B1	o.e. mark final answer Examiner's Comments Those, of the relatively few candidates, who got this correct just wrote down 2 x 3 x ¼ (e^2 – 3). Some tried to integrate g(x), with little success.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	18	

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$a = ½$	B1	or 0.5	
		i	$b = 1$	B1	<u>Examiner's Comments</u> Some candidates were able to write down the correct values of *a* and *b*. Those who chose to use transformation arguments sometimes confused the stretch (1/2 or 2) and the translation (+1 or –1). Others chose to substitute the coordinates of specific points, with variable success.	
		ii	$½\|x+1\| = \|x\|$ $\Rightarrow\ ½(x+1) = x,$	M1	o.e. ft their *a* (≠ 0), *b* (but allow recovery to correct values) or verified by subst $x = 1$, $y = 1$ into $y = ½\|x+1\|$ and $y = \|x\|$	
		ii	$\Rightarrow x = 1,\ y = 1$	A1	unsupported answers M0A0	
		ii	or $½(x + 1) = –x,$	M1	o.e., ft their *a*. *b*; or verified by subst (–1/3, 1/3) into $y = ½\|x+1\|$ and $y = \|x\|$	
		ii	$\Rightarrow x = –1/3,\ y = 1/3$	A1	or 0.33, –0.33 or better unsupported answers M0A0	
		ii	*or* $¼(x + 1)^2 = x^2$	M1	ft their *a* and *b*	
		ii	$\Rightarrow 3x^2 – 2x – 1 = 0$	M1ft	obtaining a quadratic = 0, ft their previous line, but must have an x^2 term	
		ii	$\Rightarrow x = –1/3$ or 1	A1	SC3 for (1, 1) (–1/3, 1/3) and one or more additional points	

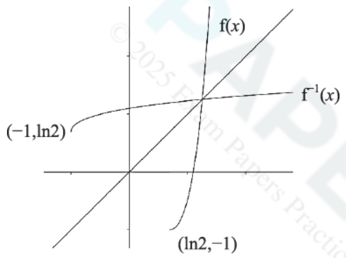
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	y = 1/3 or 1	A1	**Examiner's Comments** Most candidates, who knew what they were doing here either used ½ (x + 1) = ± x or squared both sides to find a quadratic in x. In the latter approach, some forgot to square the ½ and got the wrong quadratic. Examiners followed through their values for a and b. Some candidates omitted the y-coordinates. Candidates who found (1, 1) without showing a valid method got no marks, and there was evidence of the usual mistakes in using modulus, such as |x + 1|= |x| + 1, etc.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	$a = 2$, $b = ½$	B1B1	**Examiner's Comments** For some, this was a write-down for 2 marks. Some used transformation arguments, other substituted in particular coordinates. The most common errors were a = 3 and b = 2 or ¼ .	
		ii	$y = 2 + \cos ½\, x \leftrightarrow x \leftrightarrow y$ $x = 2 + \cos ½\, y$		(may be seen later)	
		ii	$\Rightarrow x - 2 = \cos ½\, y$	M1	subtracting [their] a from both sides (first)	need not substitute for a, b
		ii	$\Rightarrow \arccos (x - 2) = ½\, y$	M1	arccos (x –[their] a) = [their] $b \times y$	or with $x \leftrightarrow y$, need not subst for a, b
		ii	$\leftrightarrow y = f^{-1}(x) = 2 \arccos (x - 2)$	A1	cao or $2 \cos^{-1} (x - 2)$	may be implied by flow diagram
		ii	Domain $1 \le x \le 3$	M1	domain 1 to 3, range 0 to 2 π	if not stated, assume first is domain
		ii	Range $0 \le y \le 2\pi$	A1	correctly specified: must be ≤, x for domain, y or f^{-1} or $f^{-1}(x)$ for range **Examiner's Comments** We allowed plenty of follow-through marks for the first two method marks here, which were almost always gained. Fully correct inverse functions were common. As for domain and range, most seem to know that the these are the reverse of the domain and range for f. However, the 'A' mark here demanded accurate use of notation, with 'x' used for domain and 'y' for range, and this mark was often lost.	allow [1, 3], [0, 2 π] not 360° (not f)

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	7	

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	At P, $(e^x - 2)^2 - 1 = 0$			
		i	$\Rightarrow e^x - 2 = [\pm]1$,	M1	square rooting – condone no ±	
		i	$e^x = [1 \text{ or}]\ 3$			
		i	*or* $(e^x)^2 - 4e^x + 3 = 0$	M1	expanding to correct quadratic and solve by factorising or using quadratic formula	condone e^*x*^2
		i	$\Rightarrow (e^x - 1)(e^x - 3) = 0$, $e^x = 1$ or 3			
		i	$\Rightarrow x = [0 \text{ or}]\ \ln 3$	A1	*x*-coordinate of P is ln 3; must be exact **Examiner's Comments** Most candidates succeeded in finding x = ln 3, either by square rooting or solving thequadratic in e^x. The second method was somewhat compromised by setting $x = e^x$ (ratherthan a different variable) to get a quadratic in x, though we condoned this for both marks.	condone P = ln 3, but not *y* = ln 3
		ii	$f'(x) = 2(e^x - 2)e^x$	M1	chain rule	e.g. 2 *u* × their deriv of e^x
		ii		A1	correct derivative	$2(e^x - 2)x$ is M0
		ii	$= 0$ when $e^x = 2$, $x = \ln 2$ *	A1	not from wrong working **NB AG**	or verified by substitution
		ii	*or* $f(x) = e^{2x} - 4e^x + 3$	M1	expanding to 3 term quadratic with $(e^x)^2$ or e^{2x}	condone e^*x*^2
		ii	$\Rightarrow f'(x) = 2e^{2x} - 4e^x$	A1	correct derivative, not from wrong working	
		ii	$= 0$ when $2e^{2x} = 4e^x$, $e^x = 2$, $x = \ln 2$ *	A1	or $2e^x(e^x - 2) = 0 \Rightarrow e^x = 2$, $x = \ln 2$	or verified by substitution
		ii			not from wrong working **NB AG**	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = f(\ln(2)) = -1$	B1	**Examiner's Comments** This provided a simple four marks for most candidates, using a chain rule to find the derivative, setting this to zero and solving to get x = ln 2. A neat alternative method was to recognise that the $(e^x - 2)^2$ term must be non-negative and minimum when $e^x - 2 = 0$, or x = ln 2.	
		iii	$\int_0^{\ln 3}[(e^x-2)^2-1]dx$ $=\int_0^{\ln 3}[(e^x)^2-4e^x+4-1]dx$	M1	expanding brackets must have 3 terms: $(e^x)^2 - 4$ is M0, condone $e\hat{\ }x\hat{\ }2$	or if $u = e^x$, $\int_1^3[u^2 - 4u + 4 - 1]/u \, du$
		iii	$=\int_0^{\ln 3}[e^{2x}-4e^x+3]dx$	A1	$\int e^{2x} - 4e^x + 3$ [dx] (condone no dx)	$= \int u - 4 + 3/u \, du$
		iii	$=\left[\frac{1}{2}e^{2x}-4e^x+3x\right]_0^{\ln 3}$	B1	$\int e^{2x} = ½ e^{2x}$	$= [½ u^2 - 4u + 3\ln u$
		iii		A1ft	$[½ e^{2x} - 4e^x + 3x]$	
		iii	= (4.5 – 12 + 3ln3) – (0.5 – 4)			
		iii	= 3ln3 – 4 [so area = 4 – 3ln3]	A1	condone 3ln3 – 4 as final ans; mark final ans **Examiner's Comments** This proved to be a rather costly part for candidates unless they recognised the requirement to multiply out $(e^x - 2)^2 - 1$ to get $e^{2x} - 4e^x + 3$ and then integrate term-by-term. Other attempts using substitution or parts usually got nowhere. Although originally we required candidates to give the area as 4 – 3ln3, very few actually did this, so it was decided to condone a (negative) area of 3ln3 – 4.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y = (e^x - 2)^2 - 1$ $x \leftrightarrow y$			
		iv	$x = (e^y - 2)^2 - 1$			
		iv	$\Rightarrow x + 1 = (e^y - 2)^2$	M1	attempt to solve for y (might be indicated by expanding and then taking lns)	or x if x and y not interchanged yet or adding (or subtracting) 1
		iv	$\Rightarrow \pm\sqrt{(x + 1)} = ey - 2$ (+ for $y \geq \ln 2$)	A1	condone no $\pm$	
		iv	$\Rightarrow 2 + \sqrt{(x + 1)} = e^y$			
		iv	$\Rightarrow y = \ln(2 + \sqrt{(x + 1)}) = f^{-1}(x)$	A1	must have interchanged x and y in final ans	
		iv	Domain is $x \geq -1$	B1	must be $\geq$ and x (not y)	if not specified, assume first ans is domain and second range
		iv	Range is $y \geq \ln 2$	B1	or $f^{-1}(x) \geq \ln 2$, must be $\geq$ (not x or $f(x)$) if $x > -1$ and $y > \ln 2$ SCB1	
		iv		M1	recognisable attempt to reflect curve, or any part of curve, in $y = x$	$y = x$ shown indicative but not essential

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		A1	good shape, cross on $y = x$ (if shown), correct domain and range indicated. [see extra sheet for examples] **Examiner's Comments** Rather more than half of the candidates managed the inverse function well, though a few made errors at the last stage of taking the square root, and concluded with y = ln($\sqrt{(x+1)}$)+ 2, or y = ln($\sqrt{(x+1)}$+ ln 2. Some were perhaps encouraged by the previous part to multiply out $(e^x - 2)^2$ again, though they could still obtain a method mark for a step towards finding y in terms of x. It was not uncommon to see candidates taking logs of individual terms.	e.g. – 1 and ln 2 marked on axes
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$f'(x)=\dfrac{x.2(x-2)-(x-2)^2}{x^2}$	M1	quotient (or product) rule, condone sign errors only	e.g. $\dfrac{\pm x.2(x-2)\pm(x-2)^2}{x^2}$
		i		A1	correct exp, condone missing brackets here	PR: $(x-2)^2.(-x^{-2})$ + $(1/x).2(x-2)$
		i	$=\dfrac{2x^2-4x-x^2+4x-4}{x^2}$			
		i	= $(x^2-4)/x^2 = 1-4/x^2$ *	A1	simplified correctly **NB AG**	with correct use of brackets
		i	*or* $f(x) = (x^2-4x+4)/x$			
		i	= $x-4+4/x$	M1	expanding bracket and dividing each term by x correclty	must be 3 terms: $(x^2-4)/x$ is M0
		i		A1		e.g. $x-4+2/x$ is M1A0
		i	$\Rightarrow f'(x) = 1-4/x2$*	A1	not from wrong working **NB AG**	
		i	$f''(x)8/x^3$	B1	o.e. e.g. $8x^{-3}$ or $8x/x^4$	
		i	$f'(x) = 0$ when $x^2 = 4$, $x = \pm 2$	M1	$x = \pm 2$ found from $1-4/x^2 = 0$	allow for $x = -2$ unsupported
		i	so at Q, $x = -2$, $y = -8$.	A1	(–2, –8)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	f ″ (– 2) [= – 1] < 0 so maximum	B1dep	dep first B1. Can omit – 1, but if shown must be correct. Must state < 0 or negative. **Examiner's Comments** This part was very well-answered, with many getting all 7 marks. The majority of candidates opted to use the quotient rule rather than the slightly easier method of expanding the numerator and dividing through by x. Even so, provided they took care in the use of brackets, they gained the first three marks. The second part was not quite as successful. Some candidates forgot to work out the y-coordinate of Q; others got the second derivative wrong, or failing to state explicitly that for a maximum the second derivative was negative.	must use 2nd derivative test
		ii	$f(1) = (-1)^2/1 = 1$			or $(x-2)^2 = x \Rightarrow x^2 - 5x + 4 = 0$
		ii	$f(4) = (2)^2/4 = 1$	B1	verifying f(1) = 1 and f(4) = 1	$\Rightarrow (x-1)(x-4) = 0$, $x = 1, 4$
		ii	$\int_1^4 \frac{(x-2)^2}{x}\mathrm{d}x = \int_1^4 (x-4+4/x)\mathrm{d}x$	M1	expanding bracket and dividing each term by x 3 terms: $x - 4/x$ is M0	if $u = x - 2$
		ii				$\int \frac{u^2}{u+2}\mathrm{d}u = \int (u-2+\frac{4}{u+2})\mathrm{d}u$
		ii	$= \left[x^2/2 - 4x + 4\ln x\right]_1^4$	A1	$x^2/2 - 4x + 4\ln x$	$u^2/2 - 2u + 4\ln(u+2)$
		ii	= (8 – 16 + 4ln4) – (½ – 4 + 4ln1)			
		ii	= 4ln4 – 4 ½	A1cao		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Area enclosed = rectangle – curve	M1	soi	
		ii	$= 3 \times 1 - (4\ln 4 - 4\tfrac{1}{2}) = 7\tfrac{1}{2} - 4\ln 4$	A1cao	o.e. but must combine numerical terms and evaluate ln 1 – mark final ans	
		ii	*or*			
		ii	$\text{Area} = \int_1^4 [1 - \frac{(x-2)^2}{x}]\,dx$	M1	no need to have limits	
		ii	$= \int_1^4 (5 - x - 4/x)dx$	M1	expanding bracket and dividing each term by x	must be 3 terms in $(x-2)^2$ expansion
		ii	$[5x - x^2/2 - 4\ln x]_1^4$	A1	$5 - x - 4/x$	
		ii	$= 20 - 8 - 4\ln 4 - (5 - \tfrac{1}{2} - 4\ln 1)$	A1	$5x - x^2/2 - 4\ln x$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 7 ½ – 4ln4	A1cao	o.e. but must combine numerical terms and evaluate ln 1 – mark final ans **Examiner's Comments** Most candidates verified that y = 1 when x = 1 or 4, though the majority did this by re-arranging the equation $(x – 2)^2 = x$ as a quadratic and solving this. However, for the integration, many candidates failed to spot the need to expand the $(x – 2)^2$ term and divide through by x, and most attempts to use substitution (except perhaps for the somewhat fatuous u = x) or parts usually led nowhere. Those who managed this integration successfully often failed to realise they then had to subtract this value from the area of the rectangle, or did this subtraction the wrong way round. Quite a few also calculated the area of the rectangle as 4 × 1 = 4 instead of 3.	
		iii	[g(x) =] f(x + 1) – 1	M1	soi [may not be stated]	
		iii	$= \frac{(x+1-2)^2}{x+1} - 1$	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$=\frac{x^2-2x+1-x-1}{x+1}=\frac{x^2-3x}{x+1}$ *	A1	correctly simplified – not from wrong working **NB AG** **Examiner's Comments** Fewer than half of the candidates gained full marks for this part of the question. Many did not know how the translation would affect the function algebraically, for example starting with y – 1 = f(x + 1). Of those who derived a correct expression for g(x), many failed to incorporate the '-1' into their fraction.	
		iv	Area is the same as that found in part (ii)	M1	award M1 for ± ans to 8(ii) (unless zero)	
		iv	4ln4 – 7 ½	A1cao	need not justify the change of sign **Examiner's Comments** Very few candidates scored both marks here. A method mark was awarded if their answer indicated recognising the translation of their area from part (ii); but few of those who got part (ii) correct realised that the integral would be the negative of this as the transformed area is below the axis.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$1 + k$	B1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. This was an easy write-down for virtually all candidates, except those few who did not know that $e^0 = 1$.
		ii	$f'(x) = 2\,e^{2x} - 2\,k\,e^{-2x}$	B1		
		ii	$f'(x) = 0 \Rightarrow 2\,e^{2x} - 2\,k\,e^{-2x} = 0$	M1	their derivative = 0	
		ii	$\Rightarrow e^{2x} = k\,e^{-2x}$			
		ii	$\Rightarrow e^{4x} = k$, $4x = \ln k$, $x = ¼ \ln k$*	A1	NB AG	
		ii	$y = e^{(½ \ln k)} + k\,e^{(-½ \ln k)}$	M1	substituting $x = ¼ \ln k$ into $f(x)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \sqrt{k} + k/\sqrt{k} = 2\sqrt{k}$	A1cao	or $2k^{1/2}$	**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The first two marks were pretty universally earned, but deriving $x = \frac{1}{4} \ln k$, together with the final 'A' mark for getting $2\sqrt{k}$, caused a few problems, with some inaccurate logarithm work. For example, $e^{1/2 \ln k} = 1/2\ k$ was a commonly seen misconception.
		iii	$\text{Area} = \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})[\mathrm{d}x]$	B1	correct integral and limits (soi)	
		iii	$= \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})\,\mathrm{d}x = \left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]_0^{\frac{1}{2}\ln k}$	B1	$\left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]$	
		iii	$= \frac{1}{2}k - \frac{1}{2} - \frac{1}{2} + \frac{1}{2}k$	M1	$e^{\ln k} = k$ or $e^{-\ln k} = 1/k$ (soi)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= k - 1$	A1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The integration was usually correct, but, thereafter, as in part (ii), the simplification to arrive at $k - 1$ proved to be tricky, with similar errors being made.
		iv	(*A*) $g(x) = e^{2(x + \frac{1}{4}\ln k)} + ke^{-2(x + \frac{1}{4}\ln k)}$	M1	Substitute $x + \frac{1}{4}\ln k$ for x in $f(x)$	condone missing brackets
		iv	$= e^{2x}.e^{\frac{1}{2}\ln k} + k\,e^{-2x}.\,e^{-\frac{1}{2}\ln k}$	M1	$e^{p+q} = e^p \times e^q$ used	
		iv	$= (e^{\ln k})^{1/2}\,e^{2x} + k.(e^{\ln k})^{-\frac{1}{2}}\,e^{-2x}$			
		iv	$= k^{1/2}\,e^{2x} + k.k^{-1/2}.\,e^{-2x}$			

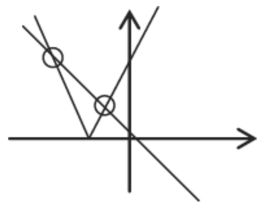
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= \sqrt{K}\,(e^{2x} + e^{-2x})$*	A1	**NB AG** – must show enough working	e.g. $k^{1/2}\, e^{2x} + k.k^{-1/2}.\, e^{-2x}$ **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. Most attempts correctly substituted $x + ¼ \ln k$ for x in f(x) to gain the first M mark, but we needed to see clear evidence of how this simplifies to the given result. Often candidates seemed to be working backwards from this without really understanding the process.
		iv	(B) $g(-x) = \sqrt{k}\,(e^{-2x} + e^{2x})$	M1	substituting $-x$ for x	condone 'f' used instead of 'g' for M1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= g(x) so g is even	A1	must include g(–x) = g(x), and either define an even function or conclude that g is even	not f(–x) = f(x) for A1 **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The definition of an even function was well known, but sometimes the structuring of the proof was indecisively presented. Some used ‘f’ instead of ‘g’ (here, f is indeed not an even function!), and we required to see either a clear statement of the definition of an even function, or a clear conclusion that g is therefore even. The structure ‘g(–x) = … = … = g(…) ⇒ g is even’ is the most transparent formulation to use in such proofs, rather than starting them by stating that g(–x) = g(x), viz the result they are trying to prove!
		iv	(C) g(x) is symmetrical about the y-axis, and	B1		
		iv	f(x) is g(x) translated ¼ ln k in x-direction	B1	allow ‘shift’ or ‘move’	or g is f translated – ¼ ln k

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	so f(x) is symmetrical about $x = ¼ \ln k$	B1	allow final B1 even if unsupported	or incorrectly supported **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The argument here proved beyond most candidates, with only 20% getting full marks. Many stated that f was an even function, perhaps thinking that any line of symmetry sufficed. Sometimes it was indeed a little difficult to decide whether candidates were referring to f or g in their answers.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	$2\cos 2y\, dy/dx = 1$	M1	$k \cos 2y\, dy/dx = 1$	or $dx/dy = k \cos 2y$, $k \cos 2y dy = dx$
		i	$\Rightarrow dy/dx = 1/(2\cos 2y)$	A1		$dy/dx = k \cos 2y$ is M0
		i	when $x = 1½$, $y = \pi/12$, $dy/dx = 1/(2\cos(\pi/6))$	M1*	substituting $y = \pi/12$ *dep 1st M1	
		i	$= 1/\sqrt{3}$	A1	or $\sqrt{3}/3$	isw from correct exact answer **Examiner's Comments** This question as also very well done, with half the candidates scoring full marks. The implicit differentiation was well understood, though there were the usual blemishes from mixing up the derivative and integral formulae for sin $2y$. A few candidates re-arranged the equation to get x in terms of y, then found dx/dy, and then the reciprocal dy/dx.
		ii	$2y = \arcsin(x - 1)$	M1		
		ii	$\Rightarrow y = ½ \arcsin(x - 1)$	A1	or $½ \sin^{-1}(x - 1)$	
		ii	translation of 1 unit in positive x-direction	B1	or translation $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	allow 'shift', but not 'move', vector only is B0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	[one-way] stretch s.f. ½ in *y*-direction	B1	not 'shrink', 'squash' etc	transformations can be in either order **Examiner's Comments** This question as also very well done, with half the candidates scoring full marks. Re-arranging the given implicit equation to give $y = \frac{1}{2}\arcsin(x-1)$ was well understood, and the transformations were usually accurately described. Note that the preferred terms here are 'translation' and 'one-way stretch'.
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24				M1	Sketch of $y = \|2x + 1\|$	condone no intercept labels, but must be a 'V' shape with vertex on –ve x axis
				A1	$y = -x$ and two intersections indicated	
			$x = -1$	B1	not from ww, condone (–1, 1)	squaring: $(2x + 1)^2 = x^2 \Rightarrow 3x^2 + 4x + 1 = 0$
			$x = -1/3$	B1	not from ww, condone (–1/3, 1/3)	$\Rightarrow (3x + 1)(x + 1) = 0$, $x = -1, -1/3$ **Examiner's Comments** Sketches of the modulus function with $y = -x$ were generally well done, though quite a few lost a mark for neither clearly indicating the intercepts nor making a clear statement that there were two of them. The roots were then usually found correctly, with less evidence of faulty modulus algebra than in recent years.
			Total	4		
25		a	$(x + 2)^2 + 3$	B1(AO1.2) B1(AO1.1) [2]	For $b = 2$ For $c = 3$ or FT their b	
		b	Since $(x + b)^2 \geq 0$, the minimum value [or minimum point on the curve] occurs when the expression in the bracket is zero	E1(AO2.2a) [1]		
			Total	3		

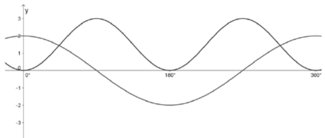
Question			Answer/Indicative content	Marks	Part marks and guidance	
26		a	Radius = 5 (2, –3)	B1(AO1.1) B1(AO1.1) [2]		
		b	$(y + 3)^2 = 21$ or $y^2 + 6y - 12 = 0$ Roots $-3+\sqrt{21}$ and $-3-\sqrt{21}$ $\left(0,-3+\sqrt{21}\right)$ and $\left(0,-3-\sqrt{21}\right)$	M1(AO1.1a) A1(AO1.1) A1(AO1.1) [3]	Substituting $x = 0$ and rearranging For one y -value All correct	
		c	$(1-2)^2 + (2+3)^2 = 1^2 + 5^2$ E.g. This is more than 25 so outside the circle	M1(AO1.1) A1(AO1.1) [2]	Or distance of (1, 2) from their centre Or distance of (1, 2) from centre is $\sqrt{26} > 5$, so outside the circle	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	Gradient CP $= \frac{1-(-3)}{-1-2} = -\frac{4}{3}$FT their C(entre) Gradient of tangent $= \frac{3}{4}$ FT their grad CP Equation of tangent $y-1=\frac{3}{4}(x-(-1))$FT their grad $4y = 3x + 7$ oe	M1(AO1.1a) M1(AO1.1) M1(AO1.1) A1(AO1.1) [4]		
			Total	11		

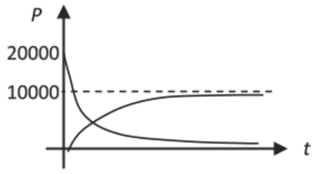
Question			Answer/Indicative content	Marks	Part marks and guidance	
27		a	Correct shape for $y=\frac{1}{x}$, translated vertically upward Crosses x-axis at $x=-\frac{1}{a}$ Asymptotes x = 0 and y = a	B1(AO1.1) B1(AO2.2a) B2(AO1.1) (AO2.2a) [4]	B1 for one asymptote correct	
		b	$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{1}{x^2}$ When x = 2 , gradient = $-\frac{1}{4}$ Gradient of normal = 4 FT their gradient $y-\frac{5}{2}=4(x-2)$FT their gradient 2y = 8x –11 oe	M1(AO1.1a) M1(AO1.1) M1(AO2.1) M1(AO1.1) A1(AO2.1) [5]	Or gradient of given line is 4 Or $y=\frac{5}{2}$ at the point on the curve where x = 2 At least one correct interim step or clear check that $\left(2, \frac{5}{2}\right)$ is on given line Any simplified form	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$2\left(\frac{1}{x}+2\right)=8x-11$ $8x^2-15x-2=0$ Other point is $x=-\frac{1}{8}$, $y=-6$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Substitution Forming quadratic, condone one error BC $x = 2$ not needed in this case		
			Total	12			
28			[1-way] stretch scale factor 2 in y-direction translation $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$	M1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]	If transformations given in reverse order then **M1, A1, M1** are still available (but not final **A1**) Or –1 in y -direction		
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
29		a	$\int \frac{dm}{m} = \int \frac{dt}{t(1+2t)}$ $\frac{1}{t(1+2t)} \equiv \frac{A}{t} + \frac{B}{1+2t}$ $\Rightarrow 1 \equiv A(1 + 2t) + Bt$ $t = 0 \Rightarrow A = 1$ $t = -½ \Rightarrow 1 = -½ B \Rightarrow B = -2$ $\Rightarrow \int \frac{dm}{m} = \int \left(\frac{1}{t} - \frac{2}{1+2t}\right) dt$ $\Rightarrow \ln m = \ln t - \ln(1 + 2t) + c$ $t = 1$, $m = 1 \Rightarrow c = \ln 3$ $\Rightarrow \ln m = \ln\left(\frac{3t}{1+2t}\right)$ $\Rightarrow m = \frac{3t}{(1 + 2t)}$	M1(AO1.1a) M1(AO3.1b) M1(AO1.1) A1A1(AO1.1 1.1) B1FT(AO2.1) M1(AO1.1) E1(AO2.1) [8]	separating variables using partial fractions substituting values, equating coeffs or cover up $A = 1$, $B = -2$ FT their *i*, *ii*, condone no *c* evaluating constant of integration AG	
		b	i $1.25 = \frac{3t}{(1+2t)}$ $\Rightarrow 1.25 + 2.5t = 3t$ $\Rightarrow t = 1.25 \div 0.5 = 2.5$ minutes	M1(AO1.1a) A1(AO1.1) [2]	Enter text here.	
		b	ii $m = \frac{3}{\left(\frac{1}{t} + 2\right)}$ $\rightarrow$1.5 [grams]	M1(AO3.1b) A1(AO2.2a) [2]	Enter text here.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		a		B1(AO1.1a) B1(AO1.1) [2]	Correct shape and symmetry for cosine graph. Correct maximum and minimum values	
		b	DR $2\cos\theta = 3\sin^2\theta$ $2\cos\theta = 3(1 - \cos^2\theta)$ $3\cos^2\theta + 2\cos\theta - 3 = 0$ $\cos\theta = \frac{-1}{3} + \frac{\sqrt{10}}{3}$ $\theta = 43.9°, 316.1°$ $\cos\theta = \frac{-1}{3} - \frac{\sqrt{10}}{3} < -1$ gives no solution	B1(AO1.2) M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO1.1) E1(AO2.4) [6]	Correct use of identity **must be seen** Rearranging to zero **must be** seen, condone one error Solve quadratic Or state that graph in part (a) only shows two solutions	

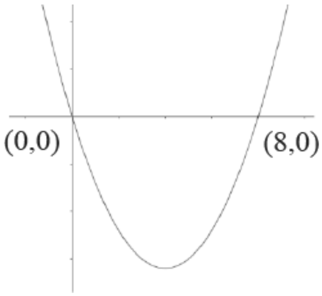
Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	8	

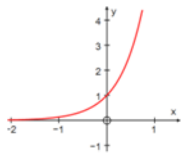
Question			Answer/Indicative content	Marks	Part marks and guidance
31		a	*A* 	M1(AO1.1) M1(AO1.1)	P_G shape through O P_R shape through (0, 20000), [condone graphs for –ve *t*]
		a	*B* asymptote for P_G = 10 000 Asymptote for P_R = 0	A1(AO2.2a) A1(AO2.2a) [4]	Or p = 10 000 Or p = 0
		b	Red squirrels zero Grey 10 000	B1(AO3.4) B1(AO3.4) [2]	
		c	One relevant comment evaluating the validity of the model	B1(AO3.5a)	E.g. One of • Grey population increases as would be expected [since grey squirrels are larger and more successful] • Red population decreases as would be expected [since red

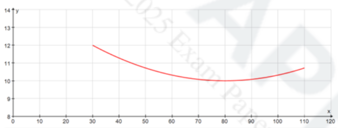
Question			Answer/Indicative content	Marks	Part marks and guidance	
				[1]	squirrels have to compete with the larger grey squirrels for food] • Number of squirrels tends to a limit as would be expected [since there is limited food and space] • Would expect grey population to grow slower at first • Would expect red population to fall slower at first	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d		M1(AO3.1b)	Attempts to differentiate either or both	
			$\frac{dP_G}{dt} = 10\ 000ke^{-kt}$	A1(AO1.1)		
			$\frac{dP_R}{dt} = -20\ 000ke^{-kt}$	A1(AO1.1)		
			so $\frac{dP_R}{dt} = -2\frac{dP_G}{dt}$	E1(AO2.1)	Or in words	
				[4]		
		e	$10\ 000\ (1 - e^{-3k}) = 20\ 000\ e^{-3k}$	M1(AO1.1a)		
			$\Rightarrow 1 - e^{-3k} = 2\ e^{-3k}$			
			$\Rightarrow e^{-3k} = \frac{1}{3}$	A1(AO1.1)		
			$\Rightarrow -3k = \ln\left(\frac{1}{3}\right)$	M1(AO1.1)	Taking natural logs of both sides	
			$\Rightarrow k = -\frac{1}{3}\ln\left(\frac{1}{3}\right) = 0.366$ or $\frac{1}{3}\ln 3$	A1cao(AO2.1)		
				[4]		
			Total	15		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32		a	P is $(\sqrt{2}, \frac{\sqrt{2}}{2})$ $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ $= \frac{\cos\theta}{-2\sin\theta}$ When $\theta = \frac{\pi}{4}$, $\frac{dy}{dx} = -\frac{1}{2}$ Equation of tangent is $(y - \frac{\sqrt{2}}{2}) = -\frac{1}{2}(x - \sqrt{2})$ $\Rightarrow \quad y = -\frac{1}{2}x + \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}$ $\Rightarrow \quad x + 2y = 2\sqrt{2}$	B1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1) B1(AO2.1) E1(AO1.1) **[6]**	oe AG	
		b	When $x = 0$, y = $\sqrt{2}$ so A is (0, $\sqrt{2}$) When $y = 0$,x= $2\sqrt{2}$ so B is ($2\sqrt{2}$, 0) Area of triangle $= \frac{1}{2}\sqrt{2} \times 2\sqrt{2} = 2$units^2	B1(AO1.1) B1(AO1.1) B1(AO1.1) **[3]**		
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
33			DR discriminant = $k^2 - 8k$ $\Rightarrow k^2 - 8k < 0$ 	B1(AO1.2) M1(AO1.1) E1(AO2.4)	Or give table of values, oe		
			$\Rightarrow k > 0$ or $k < 8$	A1(AO2.5) [4]		or $(\infty,0) \cup (8, \infty)$ or $\{k : k < 0\} \cup \{k : k > 8\}$	
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
34		a		G1(AO 1.2) G1(AO 1.2) [2]	Correct shape with x -axis as asymptote (0, 1) clearly shown	
		b	Stretch in the x-direction Stretch scale factor $\frac{1}{2}$	B1(AO 1.1a) B1(AO 1.1a) [2]	'Stretch' must be seen at least once for any marks to be awarded, but the word needn't be repeated	
		c	$\frac{dy}{dx} = 2e^{2x}$ When $x = 3$, $\frac{dy}{dx} = 2e^6$ and $y = e^6$ Tangent is $y - e^6 = 2e^6 (x - 3)$ $y = 2e^6x - 5e^6$ $y = e^6 (2x - 5)$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1a) B1(AO 1.1a) M1(AO 1.1a) A1(AO 2.1) [6]	Allow for any ke^{2x} Correct $2e^{2x}$ Allow method if 403 or better used for e^6 Must be in this form	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
35		a	$b = 80$ $c = 10$	B1(AO 1.2) B1(AO 3.3) [2]		
		b	$12 = a(30 - 80)^2 + 10$ $a = \frac{2}{50^2} = 0.0008$	M1(AO 3.3) A1(AO 1.1b) [2]	Substituting $v = 30$, $y = 12$ in the model with their values for b and c	
		c	When $v = 110$, $y = 0.0008(110 - 80)^2 + 10$ $y = 10.72$	M1(AO 3.4) A1(AO 1.1b) [2]	Substitution in the model with their values for a, b and c	
		d		G1(AO 1.1a) G1(AO 1.1a) [2]	Curve with minimum point at (80, 10) End-points (30, 12) and (110, 10.72)	Condone graph continuing outside the given interval
		e	A quadratic model giving the same value at $v = 30$ and $v = 110$ would have to have the minimum point halfway between, at $v = 70$, by symmetry This is not the case so a quadratic model is not suitable	E1(AO 3.5b) [1]	Accept equivalent statement involving symmetry, or any other sensible argument referring to relevant numerical values	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
36		a	$a = 40$ $c = 180$ $30 = 40 - b(315 - 180)^2$ $\Rightarrow b = \frac{2}{405}$ **AG**	B1(AO 3.3) B1(AO 3.3) B1(AO 2.1) [3]	At least one intermediate step needed		
		b	(*A*) 15.8	B1(AO 3.4) [1]	Allow integer answers 15 or 16		
		b	(*B*) 250 is outside the interval which provides the data and so may be extrapolating too far	E1(AO 3.5a) [1]			
		c	(*A*) $t < 90 \Rightarrow N < 0$ which is not meaningful	E1(AO 2.4a) [1]			
		c	(*B*) $t > 270$	E1(AO 3.5b) [1]			
		d	Reflection in t-axis or translated by $\begin{pmatrix}180\\0\end{pmatrix}$ or $\begin{pmatrix}-180\\0\end{pmatrix}$ Stretch in the y-direction, scale factor 40	B1(AO 1.1a) B1(AO 1.1a) [2]	Vector notation required for translation	Accept stretch in the y -direction, scale factor –40, for **B2**	
		e	$N = -40\cos t$ **or** $N = 40\cos(t - 180)$	B1(AO 3.3) [1]	oe, e.g. $N = 40\cos(t + 180)$		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		f	(*A*) Stretch graph in *t* -direction by scale factor $\frac{365}{360}$ or modify equation to $N = -40\cos\left(\frac{360}{365}t\right)$	B1(AO 3.5c) [1]	oe, e.g. $N = 40\cos\left(\frac{360}{365}t - 180\right)$; **FT** from their (**v**) for modified equations	
		f	(*B*) Maximum point moves to (182.5, 40) **or** as consistent with their answer to (*A*)	B1(AO 3.4) [1]	N.B: $N = 40\cos\left(\frac{360}{365}(t-180)\right)$ oe will 365 leave maximum point unchanged	
			Total	12		
37			DR Substitute $x = 0$ or $y = 0$ in $2x + y = 6$ Line crosses axes at (0, 6) and (3, 0) Quadratic with factor $(x – 3)$ Repeated factor $(x – 3)$ $y = \frac{2}{3}(x-3)^2$ oe	M1(AO 3.1a) A1(AO 1.1) M1(AO 3.1a) M1(AO 1.1) A1(AO 2.1) [5]	Must follow from clear reasoning	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
38		a	**DR** At intersections, $x^2 - kx = 3(k + 1) + kx - x^2$ $2x^2 - 2kx - 3(k + 1) = 0$ For touching, '$b^2 - 4ac = 0$' $4k^2 + 24(k + 1) = 0 \Rightarrow k^2 + 6k + 6 = 0$ $k = \frac{-6 \pm \sqrt{12}}{2}$ $k = -3 \pm \sqrt{3}$	M1(AO 3.1a) M1(AO 1.1) M1(AO 2.1) A1(AO 2.2a) B1(AO 1.1) B1(AO 1.1) **[6]**	Forming quadratic 0 on one side of quadratic Forming quadratic from discriminant Correct quadratic in 3-term form Use of formula or completing the square	$(k + 3)^2 = 3$	
		b	On y-axis $x = 0$, so $y = 0^2 - 0k = 0$ If they cross on the y-axis, they cross at the origin $0 = 3(k + 1) + k0 - 0^2$ $k = -1$ [so a value exists]	M1(AO 3.1a) B1(AO 2.2a) M1(AO 1.1) A1(AO 2.1) **[4]**	Use of $x = 0$ to find y Use of $x = 0$ and $y = 0$		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		a	DR $\frac{dy}{dx} = 4x^3 - 6x$ When $x = 1$, $\frac{dy}{dx} = -2$ Gradient of normal is $\frac{1}{2}$ Equation is $y = \frac{1}{2}(x-1)$	M1(AO 1.1) A1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [4]	Attempt to differentiate with at least one term correct Use of $m_1 m_2 = -1$ oe	
		b	DR Curve is part **(a)** translated up by $\frac{1}{2}$ $c = 2.5$ **Alternative method** Equation of normal is $y = \frac{1}{2}x$ Point on curve is $\left(1,\ \frac{1}{2}\right)$ $c = 2.5$	M1(AO 3.1a) M1(AO 1.1) A1(AO 2.2a) M1 M1 A1 [3]		
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
40		i	graph of cubic correct way up	B1	**B0** if stops at x-axis	must not have any ruled sections; no curving back; condone slight 'flicking out' at ends but not approaching a turning point; allow max on y-axis or in 1st or 2nd quadrants; condone some 'doubling' or 'feathering' (deleted work still may show in scans)
			crossing x-axis at –5, 1.5 and 2	B1	on graph or nearby; may be in coordinate form	allow if no graph, but marked on x-axis
			crossing y-axis at 30	B1	mark intent for intersections with both axes	condone intercepts for x and / or y given as reversed coordinates
				[3]	or $x = 0$, $y = 30$ seen if consistent with graph drawn	allow if no graph, but eg B0 for graph with intn on y-axis nowhere

Question			Answer/Indicative content	Marks	Part marks and guidance		
						near their indicated 30	
					Examiner's Comments This was completed accurately with many candidates able to sketch the graph with little preliminary working. The common errors were having a curve which stopped at the *x*-axis at one or both ends, or a curve which flicked out at an end towards a turning point, or either not marking they intercept or calling it 15. A few candidates obtained *y*-intercepts of –30, or *x*-intercepts with incorrect signs/values. In a few cases this led to a negative cubic rather than a positive cubic. Very few failed to gain any marks. Some curves were a poor shape because candidates tried to make the scales on both axes the same.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	roots of g(x) = 0 are –8, –1.5, –1	M1	or [g(x) =] (x + 1)(2x + 3)(x + 8) oe, condoning error in one bracket	NB examiners must use annotation in this part; a tick where each mark is earned is sufficient
			correct expansion of two of their two-term factors	M1	dep on attempt on f(x ±3) attempted; may be 3 or 4 terms	condone lack of brackets if correct expansions as if they were there
			correct expansion and completion to given answer, $2x^3 + 21x^2 + 43x + 24$	A1	must be working for this step before given answer	[for reference re ft: if correct, f(x – 3) = (x + 2)(2x – 9)(x – 5); allow ft if two of these brackets correct]
						or for direct expansion of all three correct factors, allow M1 for $2x^3 + 16x^2 + 2x^2 + 3x^2 + 24^x + 16x + 3x + 30$, condoning an error in one term, and A1 if no error for completion by then
				or	backwards working: allow M1 for obtaining a correct linear x a quadratic factor of given g(x)	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			finding $f(x) = 2x^3 + 3x^2 - 29x + 30$ and substituting $(x + 3)$ or $(x - 3)$ for x correct expansion for $(x + 3)^3$ and $(x + 3)^2$ correct expansion and completion to given answer, $2x^3 + 21x^2 + 43x + 24$	M1 M1 A1 **[3]**	and M1 for obtaining all 3 linear factors and A1 for justifying that these are the correct factors from using the translated roots condoning one error; condone omission of 'f(x) =' or 'y = ' **Examiner's Comments** Some candidates weren't sure which way to go so attempted pages of different combinations of brackets. Most who did know what to do got the full marks. Those who put (x+3) into the expanded f(x) gave themselves more long-winded expansions to do but most got there.	stating given answer f(x) may appear in (i) but no credit unless result is used in (ii)	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	$-16 + 84 - 86 + 24 = 6$ or $-16 + 84 - 86 + 24 - 6 = 0$	B1	or **B1** for the correct division of g(x) – 6 by (x + 2) or for the quadratic factor found by inspection, <u>and</u>, for either of these, the conclusion that no remainder means that g(–2) = 6 oe or **B1** for correct division of g(x) by (x + 2) with remainder 6 and the conclusion that g(–2) = 6 oe or **B1** for correct division of g(x) by (x + 2) with remainder 6 and the conclusion that g(–2) = 6 oe	NB examiners must use annotation in this part; a tick where each mark is earned is sufficient	
			need roots of $2x^3 + 21x^2 + 43x + 18 = 0$ soi	B1			
			attempt at division by (x + 2) as far as $2x^3 + 4x^2$ in working	M1	or clear working with g(x) and remainder of 6 found	allow working with g(x) = 0 for this M1	
			correctly obtaining $2x^2 + 17x + 9$	A1			
				M1			

Question			Answer/Indicative content	Marks	Part marks and guidance
			$\dfrac{-17 \pm \sqrt{17^2 - 4 \times 2 \times 9}}{2 \times 2}$	A1	when divided by $(x + 2)$ or $g(x) = (x + 2)(2x^2 + 17x + 9) + 6$ clearly stated at some point
			$\dfrac{-17 \pm \sqrt{217}}{4}$ oe isw	[6]	or inspection with at least two terms of three–term quadratic factor correct; if working with $g(x) = 0$ must show remainder of 6 eg in working condone one error in quadratic formula or completing square; **M0** for incorrect quadratic 'factor'
					Examiner's Comments Where candidates set out a well-organised solution, they were able to progress

Question			Answer/Indicative content	Marks	Part marks and guidance	
					directly to the fully factorised expression (with g(x) = 6 rather than 0 providing extra challenge). The majority were able to find the correct quadratic factor following division by (x + 2), with a few using synthetic division and a sizeable minority finding the solution by inspection. At this stage most then found the correct final solution, although some had difficulty squaring 17, and a few stopped at attempts to factorise. Most earned the mark for showing that g(–2) = 6 by substituting x =–2 into the equation; those who relied on their division by (x + 2) often failed to say what this showed.	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
41		i	$\left(x+\frac{1}{2}\right)^2+2\frac{3}{4}$ oe	3	**B1** for m = ½ oe **B2** for	Ignore '=0' M0 if m = 0
			$\min y = 2\frac{3}{4}$ oe or ft, isw	B1	$p=2\frac{3}{4}$oe or **M1** for 3 – their m^2	B0 if
			or showing that if y = 0, their $\left(x+\frac{1}{2}\right)^2$ is negative, so no real roots [or no solution]	[4]	ft their p, provided p > 0; ignore x value of min pt stated, even if wrong ft **B0** if only say tp rather than min, though need not justify min	explanation not 'hence' eg using $b^2 - 4ac$ on $x^2 + x +$ 3 = 0 condone B1 for $\min \text{pt} = 2\frac{3}{4}$
					Examiner's Comments Most completed the square correctly. Some candidates did not take notice of the 'hence' in the question and used the discriminant, which did not gain the final mark.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$x^2 - 4x - 12$[=0] $(x-6)(x+2)$[=0] $x = 6$ or -2 $y = 45$ or 5	M1 M1 A1 A1 **[4]**	condone one error; for equating and simplifying to solvable form for factors giving at least two terms correct, ft, or for subst in formula with at most one error ft allow **A1** for coords with x values 6 and –2 but wrong y values or **A1** each for (6, 45) and (–2, 5) **Examiner's Comments** Finding the coordinates of the points of intersection of the two curves was done well. A few forgot to work out both coordinates, and some, having found x to be 6 or –2, put (6, 0) and (–2, 0).	rearranging to zero not required if they go on to complete the square similarly for attempt at completing square	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$x^2 - 4x - (9 + k)[=0]$	M1	condone one error, but must include k **M0** if not all on one side of equation, unless completing the square	Eg allow M1 for $y = x^2 - 4x - (9 + k)$ or for $x^2 - 4x - (-9 - k)[=0]$ for completing the square method: eg $(x - 2)^2 - 4 = 9 + k$ earns M1 and the second A1, which is likely to be earned before the first A1
			$b^2 - 4ac <$ oe soi	A1	may be earned near end with correct inequality sign used there	allow '$b^2 - 4ac$ is negative' oe; 0 for just 'discriminant < 0' unless implied by later work; A0 for $\sqrt{b^2 - 4ac} < 0$;
			$16 + 4 \times (9 + k)$ (<0) oe or ft	A1	for correct substitution ft into $b2 - 4ac$; brackets / signs must be correct; allow recovery from eg wrong	for completing the square, allow A1 for $4 + 9 + k < 0$ oe
			$k < -13$ www	A1		can be earned with equality or wrong

Question			Answer/Indicative content	Marks	Part marks and guidance		
				[4]	bracket earlier but **A0** if recovery line contains additional error(s) **A0** for just $k < -52/4$	inequality, or in formula (ignore rest of formula); for completing the square method, will see $4 + 9 + k$ oe, or on opposite sides of an inequality / equation eg $9 + k < -4$ M0 for trials of values of k in $b^2 - 4ac$	
					Examiner's Comments This question posed problems for some candidates who were unsure of an appropriate strategy. However, a good number of the candidates found the set of values of k successfully. A significant minority, while knowing that they needed to use the discriminant, made arithmetical and/or algebraic errors, often caused by poor use of brackets. Few used the completing the square method.		
			Total	12			

Question			Answer/Indicative content	Marks	Part marks and guidance		
42		i	stretch parallel to y-axis oe, scale factor 2 oe	M1 A1 [2]	do not allow “squash” or “enlargement” both required **Examiner’s Comments** This caused difficulties for many. Far too many candidates did not seem to be familiar with the correct terminology, and attempted to describe what was going on by using an equation or by a (usually long-winded) sentence. “Enlargement”, “transformation” and “translation” were often seen. Similarly, a significant number of candidates ignored the request for a single transformation and described two, usually a stretch and a translation.	**M0** if two transformations described	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	translation (not "shift" or "move")	M1	if **M0 allow SC1** for eg "shift 3 units in x -direction" but not "transformation 3 units in the x-direction"	**M0** if two transformations described	
			of $\begin{pmatrix}3\\0\end{pmatrix}$, or 3 units parallel to x-axis oe	A1 [2]			
					Examiner's Comments As with part (i), many candidates opted for more general explanations. Slightly more candidates were successful with part (ii) than part (i), but once again many candidates ignored the request for a single transformation.		
			Total	4			
43			$\sqrt{1+\ln x}=0 \Rightarrow \ln x=-1$	M1(AO1.1a)			
			$x=\frac{1}{e}$	A1(AO2.2a) [2]	oe but must be exact value		
			Total	2			

Question			Answer/Indicative content	Marks	Part marks and guidance	
44		a	$x = -b$ $y = 0$	B1(AO2.2a) B1(AO1.1) [2]		
		b	For negative a, $\frac{a}{(x+b)^2} < 0$ So in this case the curve lies below the x-axis and Joe's statement is false	B1(AO2.3) E1(AO2.4) [2]	If zero scored, SC1 for $(x + b)^2$ is never negative oe	
		c	$x = -b$ is a line of symmetry $b = -2$ $3 = \frac{a}{(1+b)^2}$ a = 3	M1(AO3.1a) A1(AO2.2a) M1(AO1.1a) A1(AO1.1) [4]	Or $x = 2$ is a line of symmetry Or $3 = \frac{a}{(3+b)^2}$	Or $(1 + b)^2 = (3 + b)^3$
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
45		a	DR $\frac{dy}{dx}=0$ when $\frac{dy}{d\theta}=\cos\theta=0$	M1(AO3.1a)		
			$\theta=\frac{1}{2}\pi,\ \frac{3}{2}\pi$	M1(AO1.1)		
			$x=-2$	A1(AO1.1)		
			$y=3$	A1(AO3.2a)		
			Alternative solution			
			Maximum y occurs when $\sin\theta=1$	M1		
			$y=3$	A1		
			$\theta=\frac{1}{2}\pi$	M1		
			$x=-2$	A1		
				[4]		

Question			Answer/Indicative content	Marks	Part marks and guidance
		b	DR $7\cos\theta + 2\cos 2\theta = 0 \Rightarrow 4\cos^2\theta + 7\cos\theta - 2 = 0$	M1(AO1.1a)	Use of double angle formula
			$(4\cos\theta - 1)(\cos\theta + 2) = 0$	M1(AO1.1)	Method for solving quadratic
			$\cos\theta = \frac{1}{4}$ or -2	A1(AO1.1)	
			Reject $\cos\theta = -2$	B1(AO3.2a)	
			$\sin^2\theta = 1 - \cos^2\theta = 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}$	M1(AO3.1a)	May be seen later
			$y = 2 \pm \frac{\sqrt{15}}{4}$	A1(AO2.2a)	
				[6]	oe exact form
			Total	10	